A Channel-Fused Dense Convolutional Network for EEG-Based Emotion Recognition

Zhongke Gao, *Senior Member, IEEE*, Xinmin Wang, Yuxuan Yang, Yanli Li, Kai Ma, and Guanrong Chen, *Life Fellow, IEEE*

***Abstract*—Human emotion recognition could greatly contribute to human–computer interaction with promising applications in artificial intelligence. One of the challenges in recognition tasks is learning effective representations with stable performances from electroencephalogram (EEG) signals. In this article, we propose a novel deep-learning framework, named channel-fused dense convolutional network, for EEG-based emotion recognition. First, we use a 1-D convolution layer to receive weighted combinations of contextual features along the temporal dimension from EEG signals. Next, inspired by state-of-the-art object classification techniques, we employ 1-D dense structures to capture electrode correlations along the spatial dimension. The developed algorithm is capable of handling temporal dependencies and electrode correlations with the effective feature extraction from noisy EEG signals. Finally, we perform extensive experiments based on two popular EEG emotion datasets. Results indicate that our framework achieves prominent average accuracies of 90.63% and 92.58% on the SEED and DEAP datasets, respectively, which both receive better performances than most of the compared studies. The novel model provides an interpretable solution with excellent generalization capacity for broader EEG-based classification tasks.**

***Index Terms*—Brain–computer interface (BCI), convolutional neural network (CNN), deep learning (DL), electroencephalogram (EEG), emotion recognition.**

I. Introduction

HUMAN emotion has a great impact on our daily activities, which is concerned with various actions, such as relaxation, work, and entertainment. There are increasing research interests in the relationship between emotions and physiological functions [1]. It has been found that positive emotions could reflect pleasurable engagement, and are beneficial for human health and attitude [2]. However, accompanied with complaints of physical symptoms, negative emotions may adversely influence mental health and even cause serious psychological problems [3]. As the information explosion through social channels, it is quite challenging to reveal one's emotional clues. Recently, affective computing (AC) has emerged under the demand of deep knowledge and reasonable utilization for emotion [4], [5]. It is a promising area of research that has attracted increasing attentions from numerous cross-curricular fields, ranging from neuroscience to computer engineering. Emotion would be subtly influenced by multiple external and psychological factors, and is a combination of time, space, experience, and cultural background [6]. This aggravates the difficulties for emotion recognition research. Although great efforts have been made to explore the mechanisms and methods for emotion recognition [7], due to the intricate external patterns, effective emotion recognition methods are still in high demand for many technological applications.

There are several signal clues that are recorded to evaluate emotional states, such as facial expressions [8], speech signals [9], text messages [10], and physiological indexes like electrocardiogram (ECG) [11], electroencephalogram (EEG) [12]–[14], and dermal resistance [15]. The detection approaches, which merge facial expressions, speech signals, text messages, and other nonphysiological sources together, are fairly economical. However, these clues are varying from different human living habits and cultural backgrounds, thus not reliable to a certain degree. During collecting facial expressions, face images should be taken with high quality in good conditions. One may deliberately conceal the true feelings to trick the camera and the computer. Though these problems do not exist in some physiological indexes like dermal resistance, they could be affected by such factors as humidity and temperature, therefore they do not operate satisfactorily.

Among these clues, EEG is the most preferred source in emotion recognition research due to its good temporal resolution and information richness. Compared with other sources, EEG signal is more effective and authentic with the property of a strong unforgeability. Many studies have proved the correlations between emotional states and EEG signals in different brain regions [16]. Moreover, with the development and popularization of wearable EEG devices and dry electrode techniques [17], [18], EEG online systems can be easily implemented and are promising for practical applications in various tasks, such as sleep stage scoring [19], disease detection [20], and driver fatigue evaluation [21]. Therefore, EEG signal is

Manuscript received January 10, 2020; revised February 14, 2020; accepted February 20, 2020. Date of publication February 25, 2020; date of current version December 10, 2021. This work was supported in part by the National Natural Science Foundation of China under Grant 61922062 and Grant 61873181, and in part by the Hong Kong Research Grants Council through GRF under Grant CityU-11200317. *(Corresponding author: Zhongke Gao.)*

Zhongke Gao, Xinmin Wang, Yuxuan Yang, and Yanli Li are with the School of Electrical and Information Engineering, Tianjin University, Tianjin 300072, China (e-mail: zhongkegao@tju.edu.cn).

Kai Ma is with the Tencent Jarvis Lab, Shenzhen 518057, China (e-mail: kylekma@tencent.com).

Guanrong Chen is with the Department of Electronic Engineering, City University of Hong Kong, Hong Kong (e-mail: eegchen@cityu.edu.hk).

Color versions of one or more figures in this article are available at https://doi.org/10.1109/TCDS.2020.2976112.

Digital Object Identifier 10.1109/TCDS.2020.2976112

用于脑电图情绪识别的通道融合密集卷积网络

高钟科，高级会员，IEEE，王新敏，杨宇轩，李艳丽，马凯，以及陈冠荣，终身会士，IEEE

摘要—人类情感识别能够极大地促进人机交互，在人工智能领域具有广阔的应用前景。识别任务中的一个挑战是从脑电图（EEG）信号中学习有效的表示并保持稳定的性能。在本文中，我们提出了一种名为通道融合密集卷积网络的深度学习框架，用于基于 EEG 的情感识别。首先，我们使用一维卷积层接收来自 EEG 信号在时间维度上的加权组合的上下文特征。接下来，受最先进的物体分类技术的启发，我们采用一维密集结构来捕获电极在空间维度上的相关性。所开发的算法能够处理时间依赖性和电极相关性，并从噪声 EEG 信号中提取有效特征。最后，我们在两个流行的 EEG 情感数据集上进行了广泛的实验。结果表明，我们的框架在 SEED 和 DEAP 数据集上分别实现了 90.63%和 92.58%的显著平均准确率，这两个数据集的性能均优于大多数对比研究。 新型模型为更广泛的基于脑电图（EEG）的分类任务提供了一种具有出色泛化能力的可解释解决方案。

关键词—脑机接口（BCI）、卷积神经网络（CNN）、深度学习（DL）、脑电图（EEG）、情绪识别。

I. I

人类情绪对我们的日常活动有重大影响，这涉及到各种行为，如放松、工作和娱乐。人们越来越关注情绪与生理功能之间的关系[1]。研究发现，积极情绪可以反映愉悦的参与，对人类健康和态度有益[2]。然而，伴随着身体症状的抱怨，消极情绪这种情况可能对心理健康产生不利影响，甚至导致严重的心理问题[3]。随着通过社交渠道的信息爆炸，揭示一个人的情感线索非常具有挑战性。近年来，在深度知识和合理利用情感的需求下，情感计算（AC）应运而生[4]，[5]。这是一个吸引了众多跨学科领域关注的、充满前景的研究领域，从神经科学到计算机工程都有涉及。情感会受到多种外部和心理因素的影响，是时间、空间、经验和文化背景的综合体现[6]。这加剧了情感识别研究的难度。尽管人们在探索情感识别的机制和方法方面付出了巨大努力[7]，但由于外部模式的复杂性，有效的情感识别方法对于许多技术应用仍然需求迫切。

有几种信号线索用于评估情绪状态，例如面部表情[8]、语音信号[9]、短信[10]以及生理指标，如心电图（ECG）[11]、脑电图（EEG）[12]–[14]和皮肤电阻[15]。将面部表情、语音信号、短信和其他非生理来源结合起来的检测方法相当经济。然而，这些线索因不同的人类生活习惯和文化背景而有所不同，因此在一定程度上不可靠。在收集面部表情时，应在良好条件下拍摄高质量的面部图像。人们可能会故意隐藏真实感受以欺骗摄像头和计算机。尽管这些问题在某些生理指标（如皮肤电阻）中不存在，但它们可能受到湿度、温度等因素的影响，因此不能令人满意。

在这些线索中，脑电图（EEG）因其良好的时间分辨率和信息丰富性，成为情绪识别研究中最偏好的来源。与其他来源相比，脑电图信号具有更强的不可伪造性，因此更为有效和真实。许多研究已经证明了不同脑区情绪状态与脑电图信号之间的相关性[16]。此外，随着可穿戴脑电图设备和干电极技术的开发与普及[17][18]，脑电图在线系统可以轻易实现，并在睡眠阶段评分[19]、疾病检测[20]和驾驶员疲劳评估[21]等多种任务中展现出良好的应用前景。因此，脑电图信号是

稿件收到日期为 2020 年 1 月 10 日；修改日期为 2020 年 2 月 14 日；接受日期为 2020 年 2 月 20 日。发表日期为 2020 年 2 月 25 日；当前版本日期为 2021 年 12 月 10 日。本研究部分由国家自然科学基金（项目编号 61922062 和 61873181）资助，部分由香港研究资助局通过一般研究基金（项目编号 CityU-11200317）资助。（通讯作者：高钟科）。高钟科、王新敏、杨宇轩和李艳丽为天津大学电气与信息工程学院，中国天津 300072（邮箱：zhongkegao@tju.edu.cn）。马凯为腾讯 Jarvis 实验室，中国深圳 518057（邮箱：kylekma@tencent.com）。陈冠荣为香港城市大学电子工程系，香港（邮箱：eegchen@cityu.edu.hk）。本文中一个或多个图的颜色版本可在 https://doi.org/10.1109/TCDS.2020.2976112 获取。数字对象标识符: 10.1109/TCDS.2020.2976112

selected as the source for our emotion recognition study in this article.

It is noted that the recorded EEG signals are inevitably mixed with noises due to their low signal-to-noise ratios (SNRs), which makes it quite challenging to design computational algorithms for recognizing emotions. Actually, EEG signals are time-varying data, and recorded from multiple electrodes that are arranged with the standard 10–20 system [22]. These task-related signals contain the rich spatial–temporal information, which can reflect electrode correlations across the spatial dimension and contextual dependencies across the temporal dimension, respectively. The effective extraction of spatial–temporal information can help better recognize the emotions. Various methods have been developed to handle the temporal dependencies from EEG signals, including time-frequency analysis [23], complex network methods [24]–[27], and nonlinear analysis [28], [29]. Deserved to be mentioned, differential entropy (DE) and power spectral density (PSD) features have been proven valid for emotion recognition [12], [30], [31]. Meanwhile, principal component analysis (PCA) [32] and Fisher transform [33] are commonly used for feature selection and optimization. However, most of these feature-based methods mainly focus on extracting temporal features across a single channel while neglecting the information of electrode correlations. Additionally, the extraction of some features is quite time consuming, especially for nonlinear analysis, which cannot meet most demands online.

In recent years, deep-learning (DL) techniques have been developed rapidly, drawing a great deal of attentions in diverse research fields. Various network architectures have been proposed, such as deep belief networks (DBNs) [34], convolutional neural networks (CNNs) [35], and recurrent neural networks (RNNs) [36]. These models are superior in computational efficiency and model performance, and have exerted prodigious impacts in many fields, such as image classification [37], speech recognition [38], and time-series prediction [39], [40]. Note that EEG signals are of great importance in the extraction of spatial correlations and temporal dependencies, which slightly differs from 2-D images and speech signals. There are considerable explorations on EEG-based classification tasks, including motor imagery classification [41]–[43], fatigue driving evaluation [44], [45] and emotion recognition [46]–[48]. There is a detailed survey about CNNs in EEG analysis [49]. These works greatly enrich the explorations of DL-based EEG signal analyses. However, some challenges still remain to be solved. Most of the frameworks for EEG analysis are shallow networks, simply containing convolutional layers, pooling layers, and fully connected layers. This results in lacking the capability of nonlinear approximation, which may correspond to uncompetitive effects.

To address the above problems, we propose a novel deep architecture in this article, named channel-fused dense convolutional network (CDCN), for EEG-based emotion recognition tasks. This article has two contributions as follows.

1) We propose a CDCN framework to deal with contextual information and spatial correlations for emotion recognition, where the 1-D convolution and dense connections are integrated to enhance the performance. This method guarantees better performances compared with others competitive recognition methods for EEG-based emotion recognition on the SEED and DEAP datasets.
2) CDCN could elegantly handle the problems that shallow networks have weak capacity to deal with, where 2-D convolution requires more information of temporal dependencies and electrode correlations. In CDCN, the initial 1-D convolution layer plays the role of temporal feature selections, which outperforms other compared methods regarding both efficiency and accuracy, and could extend the application to broader EEG-based classification tasks.

To study the emotion recognition problem, the layout of this article is organized as follows. Section II briefly introduces the related work in emotion feature methods and DL techniques applied in EEG analyses. Section III presents the developed framework on emotion recognition tasks. Section IV reports the experimental results evaluated on two popular emotion datasets. Section V discusses the possibility of improvements for emotion recognition tasks. Section VI presents the conclusions.

II. Related Work

A. Emotion Features

EEG signal has been one of the most popular signal forms in emotion recognition tasks, which has attracted lots of attention and many feature methods have explored. The common features can be mainly associated with the following three categories: 1) time-domain features; 2) frequency-domain features; and 3) functional connectivity features [50].

Time-domain features aim to extract the temporal information through EEG signals, such as statistical features and fractal dimension features. The work [51] extracted statistical features, fractal dimension features, and other features of EEG signals as the inputs of support vector machine (SVM) for binary valence-arousal (VA) recognition on the DEAP dataset, which reached an average accuracy of 73.10%. Frequency-domain features mainly capture the spectral information from EEG signals, such as band power and DE [52], [53]. In [54], DE features were extracted as the inputs of a graph regularized extreme learning machine classifier, which received a mean accuracy of 91.07% on the SEED dataset. In particular, the DE feature is receiving increasing attention in emotion analysis tasks. Then, functional connectivity features focus on the correlation and synchronization information between sensors pairs, which serves as the spatial information of EEG signals. Moreover, there is a detailed survey of emotion recognition methods [55], which reviewed the main aspects involved in the recognition process, including subjects, studied features, and popular classifiers.

B. Deep Learning for EEG Analysis

Many methods have been proposed to investigate proper computational models for emotion recognition using EEG signals. Various DL architectures have been applied in classifying EEG signals to solve different recognition tasks. Typically,

被选为本篇文章情绪识别研究的来源。

值得注意的是，由于脑电图（EEG）信号的信噪比（SNR）较低，记录的 EEG 信号不可避免地会混入噪声，这使得设计用于识别情绪的计算算法变得相当具有挑战性。实际上，EEG 信号是时变数据，并从按照标准 10-20 系统排列的多个电极上记录[22]。这些与任务相关的信号包含了丰富的时空信息，这些信息可以分别反映电极在空间维度上的相关性以及时间维度上的上下文依赖性。有效提取时空信息有助于更好地识别情绪。为了处理 EEG 信号中的时间依赖性，已经开发出多种方法，包括时频分析 [23]、复杂网络方法 [24]-[27] 和非线性分析 [28], [29]。值得提及的是，微分熵（DE）和功率谱密度（PSD）特征已被证明对情绪识别是有效的 [12], [30], [31]。同时，主成分分析（PCA）[32] 和 Fisher 变换 [33] 常用于特征选择和优化。 然而，这些基于特征的方法主要集中于从单个通道提取时间特征，而忽略了电极间相关性的信息。此外，某些特征的提取非常耗时，尤其是非线性分析，这无法满足大多数在线需求。近年来，深度学习（DL）技术发展迅速，在各个研究领域引起了广泛关注。提出了多种网络架构，如深度信念网络（DBNs）[34]、卷积神经网络（CNNs）[35]和循环神经网络（RNNs）[36]。这些模型在计算效率和模型性能方面具有优势，并在许多领域产生了巨大影响，如图像分类[37]、语音识别[38]和时间序列预测[39]、[40]。需要注意的是，脑电图（EEG）信号在提取空间相关性和时间依赖性方面具有重要意义，这与二维图像和语音信号略有不同。 在基于脑电图（EEG）的分类任务中，已有相当多的探索，包括运动想象分类[41]–[43]、疲劳驾驶评估[44]，[45]和情绪识别[46]–[48]。关于 CNN 在 EEG 分析中的应用有详细的综述[49]。这些工作极大地丰富了基于深度学习（DL）的 EEG 信号分析的探索。然而，仍有一些挑战有待解决。大多数 EEG 分析的框架是浅层网络，仅包含卷积层、池化层和全连接层。这导致缺乏非线性逼近能力，可能对应于效果不具竞争力的表现。

为解决上述问题，本文提出了一种新的深度架构，命名为通道融合密集卷积网络（CDCN），用于基于 EEG 的情绪识别任务。本文有两个主要贡献如下。

1) 我们提出了一个 CDCN 框架，用于处理情绪识别中的上下文信息和空间相关性在 1-D 卷积和密集连接集成以提升性能的地方。与 SEED 和 DEAP 数据集上的其他竞争性识别方法相比，该方法保证了更好的性能。

2) CDCN 可以优雅地处理浅层网络处理能力较弱的问题，其中 2-D 卷积需要更多关于时间依赖性和电极相关性的信息。在 CDCN 中，初始的 1-D 卷积层起着时间特征选择的作用，在效率和准确性方面都优于其他比较方法，并可将应用扩展到更广泛的基于脑电图的情绪分类任务。

为了研究情绪识别问题，本文的布局组织如下。第二节简要介绍了情绪特征方法和应用于脑电图分析中的深度学习技术。第三节介绍了开发中的情绪识别任务框架。第四节报告了在两个流行情绪数据集上的实验结果。第五节讨论了情绪识别任务改进的可能性。第六节总结了结论。

II. R W

A. 情感特征

脑电图（EEG）信号一直是情感识别任务中最受欢迎的信号形式之一，吸引了大量关注，许多特征方法都进行了探索。常见特征主要可分为以下三类：1）时域特征；2）频域特征；3）功能连接特征[50]。

时域特征旨在通过脑电图（EEG）信号提取时间信息，例如统计特征和分形维数特征。文献[51]提取了 EEG 信号的统计特征、分形维数特征等，作为支持向量机（SVM）的输入，在 DEAP 数据集上进行二元效价-唤醒（VA）识别，平均准确率达到 73.10%。频域特征主要捕捉 EEG 信号的频谱信息，例如频带功率和 DE[52][53]。在文献[54]中，DE 特征被提取作为图正则化极限学习机分类器的输入，在 SEED 数据集上获得平均准确率 91.07%。特别地，DE 特征在情绪分析任务中正获得越来越多的关注。此外，功能连接特征关注传感器对之间的相关性和同步信息，这构成了 EEG 信号的空间信息。此外，文献[55]对情绪识别方法进行了详细综述，回顾了识别过程中的主要方面，包括受试者、研究特征和流行分类器。

B. 脑电图分析的深度学习

针对使用脑电图信号进行情绪识别的适当计算模型，已经提出了许多方法。各种深度学习架构已被应用于分类脑电图信号以解决不同的识别任务。通常，

most of the existing DL-based EEG studies can be summarized into two categories. The first one is based on using EEG signals as the input of a network. The second one is based on using the extracted features from EEG signals as the input of a network.

In EEG analysis tasks, it requires the developed model able to capture internal information from EEG signals. In [56], a compact fully convolutional network EEGNet was proposed to process EEG signals, which performed effectively in four different brain–computer interface (BCI) classification tasks. In our previous work [45], a spatial–temporal CNN was developed to deal with EEG signals in the fatigue detection task. RNNs are capable of extracting temporal information in EEG analyses. In [57], the model was based on a convolutional long short-term memory and a new temporal margin-based loss function, which achieved overall accuracies of 78.72% and 79.03% for recognizing valence and arousal emotions, respectively, on the DEAP dataset. A long short-term memory architecture was developed in [58] for cognitive workload estimation with accuracy up to 93%. These studies have proved that DL methods can learn effective representations from EEG signals.

Another direction toward better performance is to combine analysis with prior knowledge. In [59], EEG signals were transformed into multispectral images and a recurrent convolutional network was trained for mental load evaluation. EEG sequences were converted into 2-D graph matrixes with spectral filtering and dynamical graph CNNs were introduced for EEG emotion recognition in [60]. A hierarchical CNN was trained in [61] with 2-D maps organized from DE features in all channels, which was found efficient in emotion recognition tasks. A spatial–temporal RNN was proposed to integrate the feature learning from both spatial and temporal dimensions of signal sequences in [62], where the final accuracy for emotion recognition reached 89.5% on the SEED dataset. The above studies combined with prior knowledge to provide a good approach, which allows building specific frameworks for EEG-based classification tasks.

Although DL methods have achieved great progresses in EEG recognition tasks, there are still many challenges to be solved. For instance, the existing feature-based DL methods may attach little importance on the information of electrode correlations. To solve these problems, we extract the DE features from EEG signals, and then convert them into 2D matrixes as the input of the CDCN model, which allows effective processing of temporal dependencies and electrode correlations with series-specific modifications for EEG-based emotion recognition.

III. Methods

In this section, we first introduce the DE feature and use it as the input of the CDCN model, then present the developed framework with the model architecture and implementation details.

A. DE Feature

Various methods have been used to extract vital features from EEG signals. In the works reviewed, DE features have showed great abilities to measure the complexity of continuous random variables in emotion recognition tasks [31]. The calculation formula of DE feature is defined as

$$h(X) = -\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\frac{(x-\mu)^2}{2\sigma^2} \log\frac{1}{\sqrt{2\pi\sigma^2}} \exp\frac{(x-\mu)^2}{2\sigma^2} dx = \frac{1}{2}\log 2e\pi\sigma^2 \tag{1}$$

where X follows the Gaussian distribution $N(\mu, \sigma^2)$, π, and e are constants, and x is a variable. DE feature has been proven to be equivalent to the logarithmic PSD for fixed-length EEG segments in a certain frequency band. So, we extract the DE features in five main frequency bands (delta: 1–3 Hz, theta: 4–7 Hz, alpha: 8–13 Hz, beta: 14–30 Hz, and gamma: 31–50 Hz) for each channel. If the EEG signal has E channels, each feature matrix has the dimensions of $[E, 5]$, which serves as the input of the CDCN model.

B. Model Architecture

DenseNet was proposed by Huang *et al.* [63], and compared with traditional CNNs, it has several compelling advantages: strengthen the feature propagation, alleviate the vanishing gradient problem, encourage feature reuse, and substantially reduce the number of parameters. However, DenseNet and its extensions are specially designed for 2-D image recognition tasks. Their structures for image processing actually contain 2-D convolutions and pooling operations, which are improper for multichannel EEG processing. Imposing the patterns in applications, it may lead to the information loss of electrode correlations and temporal dependencies in EEG signals. Taking the above issues into consideration, the proposed CDCN framework uses the 1-D convolution to deal with temporal information and the modified dense blocks to emphasize the importance of electrode correlations. Fig. 1 shows the pipeline of the proposed method for EEG-based emotion recognition.

Let matrix $X \in R^{E\times F}$ denote input samples with matching labels, where each signal sample has E electrodes and each electrode has F features, and C denotes the categories of emotional states. On the SEED dataset, the dimension of the input samples is 62-by-5; on the DEAP dataset, the dimension of the input samples is 32-by-5. The proposed CDCN contains convolutional layers, pooling layers, dense blocks, and transition blocks.

The dense block is used to improve the information flow and strengthen the feature reuse. It introduces direct connections from any layer to the subsequent layers, which is adopted to extract high-level features through the feature maps from the previous layer. Let x_l be the lth layer's output and the $(l+1)$th layer's input in the traditional neural network, which can be expressed as

$$x_l = H_l(x_{l-1}) \tag{2}$$

where H is the nonlinear transformation in the lth layer. Hence, for the lth layer in the dense block, it receives the feature maps from all preceding layers. Thus

$$x_l = H_l([x_0, x_1, \ldots, x_{l-1}]) \tag{3}$$

目前基于深度学习的脑电图（EEG）研究大多可分为两类。第一类是基于将脑电图信号作为网络输入。第二类是基于将从脑电图信号中提取的特征作为网络输入。

在脑电图分析任务中，需要开发的模型能够从脑电图信号中捕获内部信息。在[56]中，提出了一种紧凑的全卷积网络 EEGNet 来处理脑电图信号，该网络在四种不同的脑机接口（BCI）分类任务中表现有效。在我们的先前工作[45]中，开发了一种时空卷积神经网络来处理疲劳检测任务中的脑电图信号。循环神经网络能够从脑电图分析中提取时间信息。在[57]中，模型基于卷积长短期记忆网络和一个新的基于时间边界的损失函数，在 DEAP 数据集上识别效价和唤醒情绪的总体准确率分别达到了 78.72%和 79.03%。在[58]中，开发了一种长短期记忆架构用于认知负荷估计，准确率高达 93%。这些研究表明深度学习方法能够从脑电图信号中学习有效的表示。

另一个提升性能的方向是将分析与先验知识相结合。在[59]中，脑电图信号被转换为多光谱图像，并训练了一个循环卷积网络用于心理负荷评估。脑电图序列通过频谱滤波转换为二维图矩阵，并在[60]中引入了动态图卷积网络进行脑电图情绪识别。在[61]中，使用从所有通道的 DE 特征组织成的二维图训练了一个层次化卷积网络，该网络在情绪识别任务中被发现非常有效。在[62]中，提出了一种时空循环神经网络，用于整合信号序列在空间和时间维度上的特征学习，最终在 SEED 数据集上达到 89.5%的情绪识别准确率。上述研究结合先验知识，提供了一种良好方法，允许为基于脑电图的分类任务构建特定框架。

尽管深度学习方法在脑电图识别任务中取得了巨大进展，但仍有许多挑战需要解决。例如，现有的基于特征的深度学习方法可能不太重视电极相关性的信息。为了解决这些问题，我们从脑电图信号中提取 DE 特征，然后将它们转换为 2D 矩阵作为 CDCN 模型的输入，这使得模型能够有效处理时间依赖性和电极相关性，并对脑电图识别进行特定于序列的修改。

III. M 在本节中，我们首先介绍 DE 特征，并将其作为 CDCN 模型的输入，然后介绍开发的框架，包括模型架构和实现细节。

A. DE 特征

已经使用各种方法从脑电图信号中提取重要特征。在回顾的文献中，DE 特征有显示了在情绪识别任务中测量连续随机变量复杂度的强大能力[31]。DE 特征的计算公式定义为

$$h(X) = -\int_{-\infty} \frac{1}{\sqrt{2\pi\sigma}} \exp\left(\frac{(x-\mu)}{2\sigma}\right) \log \frac{1}{\sqrt{2\pi\sigma}} \exp\left(\frac{(x-\mu)}{2\sigma}\right) dx = \frac{1}{2}\log 2e\pi\sigma \tag{1}$$

X 服从高斯分布 N(μ, σ)，π和 e 是常数，x 是变量。DE 特征已被证明在特定频段内固定长度的 EEG 片段中与对数 PSD 等效。因此，我们为每个通道在五个主要频段（δ：1-3 Hz，θ：4-7 Hz，α：8-13 Hz，β：14-30 Hz，和γ：31-50 Hz）中提取 DE 特征。如果 EEG 信号有 E 个通道，每个特征矩阵的维度为[E, 5]，这作为 CDCN 模型的输入。

B. 模型架构

DenseNet 由 Huang 等人[63]提出，与传统 CNN 相比，它具有几个显著优势：增强特征传播、缓解梯度消失问题、鼓励特征重用，并大幅减少参数数量。然而，DenseNet 及其扩展专门为二维图像识别任务设计。它们用于图像处理的架构实际上包含二维卷积和池化操作，这些操作不适用于多通道脑电图（EEG）处理。将这种模式应用于实际场景，可能会导致脑电信号中电极相关性和时间依赖性的信息丢失。考虑到上述问题，所提出的 CDCN 框架使用一维卷积处理时间信息，并采用改进的密集块来强调电极相关性的重要性。图 1 展示了所提出方法在基于 EEG 的情绪识别中的流程。

让矩阵 X ∈ R 表示具有匹配标签的输入样本，其中每个信号样本有 E 个电极，每个电极有 F 个特征，C 表示情绪状态的类别。在 SEED 数据集上，输入样本的维度是 62-by-5；在 DEAP 数据集上，输入样本的维度是 32-by-5。所提出的 CDCN 包含卷积层、池化层、密集块和过渡块。

密集块用于提高信息流并加强特征重用。它引入了从任何层到后续层的直接连接，用于通过前一层的特征图提取高级特征。在传统神经网络中，设 x 为第 l 层的输出和第(l+1)层的输入，可以表示为

(2) 其中 H 是第 l 层的非线性(x)变换。因此，对于密集块中的第 l 层，它接收所有前一层传来的特征图。因此

$$x = H([x, x, \ldots, x]) \tag{3}$$

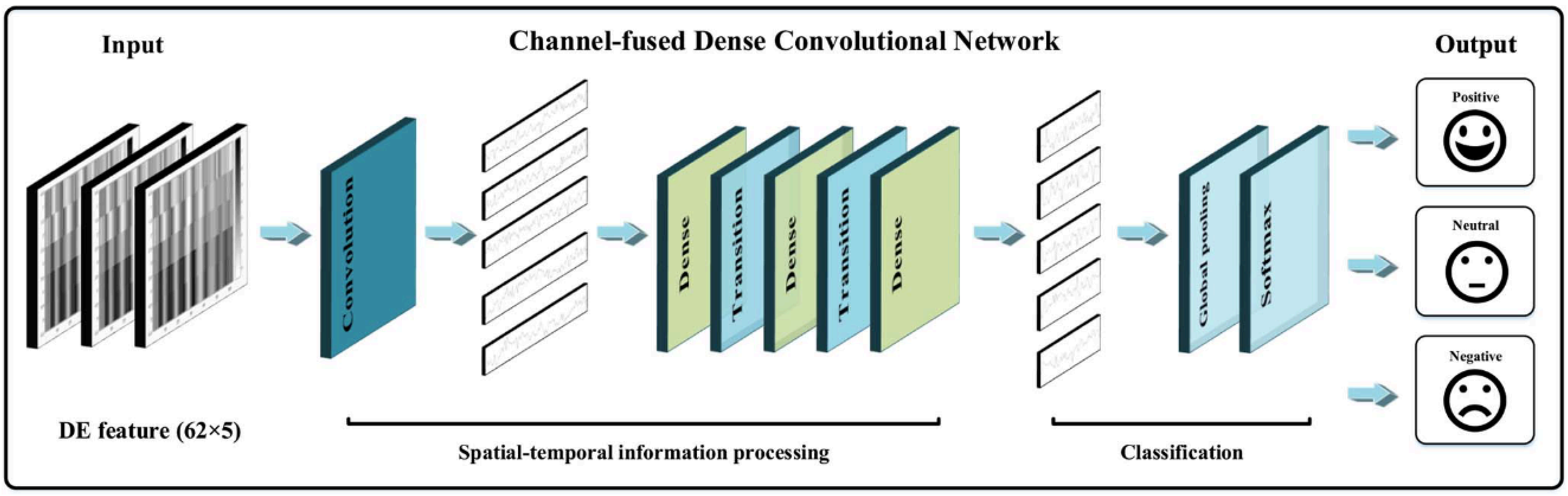


Fig. 1. Pipeline of the proposed method for EEG-based emotion recognition.

TABLE I
DETAILS OF THE DEVELOPED CDCN. THE SECOND COLUMN DENOTES THE OUTPUT SIZE OF THE CURRENT LAYER. THE SYMBOL [] CONTAINS THE KERNEL SIZE, THE NUMBER OF FEATURE MAPS AND THE TYPE OF LAYER, RESPECTIVELY

Layers	Output size	CDCN (k=12)
Input	62 × 5	–
Convolution	62 × 1	[1×5, map 24, stride 1, conv]
Dense block 1	62 × 1	[3×1, map 36: 96, conv] ×6
Transition block 1	62 × 1	[1×1, map 96, conv]
	31 × 1	[2×1, stride 2, maxpooling]
Dense block 2	31 × 1	[3×1, map 108:168, conv] ×6
Transition block 2	31 × 1	[1×1, map 168, conv]
	16 × 1	[2×1, stride 2, maxpooling]
Dense block 3	16 × 1	[3×1, map 180:240, conv] ×6
Classification	240	global average pooling
	3	softmax

where $[x_0, x_1, \ldots, x_{l-1}]$ denotes the cascade operation of $(l-1)$ layers' feature maps.

The transition block is employed to reduce the size of input feature maps, which needs some modifications to work for EEG signals. For practical use, the transition block consists of a batch normalization layer, a convolutional layer, and a max-pooling layer.

In our experiment, the depth of the CDCN is selected to 24 layers with details presented in Table I. The input dimensions and the number of emotions mentioned here corresponds to the SEED dataset. The grow rate k and the number of dense blocks are set to 12 and 3, respectively. Each dense block equally contains six convolution blocks, which orderly consists of a batch normalization [64], a rectified linear activation (ReLU) [65], and a 3×1 convolution with zero-padding to keep the size of feature maps fixed. In this case, the transition block is composed of a 1×1 convolution and a 2×1 max-pooling with stride 2.

In the framework, the $1 \times F$ convolution layer is initially performed as the input layer with twice the grow rate k feature maps, where N denotes the number of features per electrode. Three dense blocks are joined together. Meanwhile, two transition blocks are inserted between two contiguous dense blocks. A global average pooling is employed, followed by the last dense block, and then a Softmax classifier is appended. In the three dense blocks, the sizes of feature maps are 62×1, 31×1, and 16×1, respectively. If each convolution in the dense block produces k feature maps, the l th layer has $k_0 + k \times l$ output feature maps, where k_0 denotes the number of the layer's feature maps before the dense block. For the convolution layer in each transition block, the number of feature maps is the same as the previous convolution layer.

C. Implementation Details

The backpropagation through time (BPTT) algorithm is used to optimize the network parameters until receiving the optimal solutions or reaching the maximum of epochs. Besides, the cross-entropy objective function is employed as the loss function for model optimization [66], which is expressed as

$$\text{Loss} = \text{cross\_entropy}\big(y, y^p\big) \tag{4}$$

where y and y^p denote the ground truth of train samples and the predicted ones, respectively. During the training process, the function *Loss* is aimed at reducing the loss value to improve the prediction accuracy of the model. We train the CDCN framework using the Adam algorithm [67] with a learning rate of 10^{-3}. The batch size is set to 64. The ground truth of the validation set is coded by one-hot states. The model is implemented with the Keras library,[1] which is extended from Google Tensorflow.[2] Then, we save the best model by monitoring on the validation set, and the elapsed time for model training with a GeForce Titan X for about twenty minutes.

IV. EXPERIMENTS

In this section, we first introduce the SEED dataset and provide a detailed explanation of dataset preprocessing, which is used for evaluating the performance of the proposed method. Then, we analyze the model performance on the SEED dataset and show performance comparisons with other competitive methods used previously for similar researches. Finally,

[1] https://keras.io/
[2] https://www.tensorflow.org/

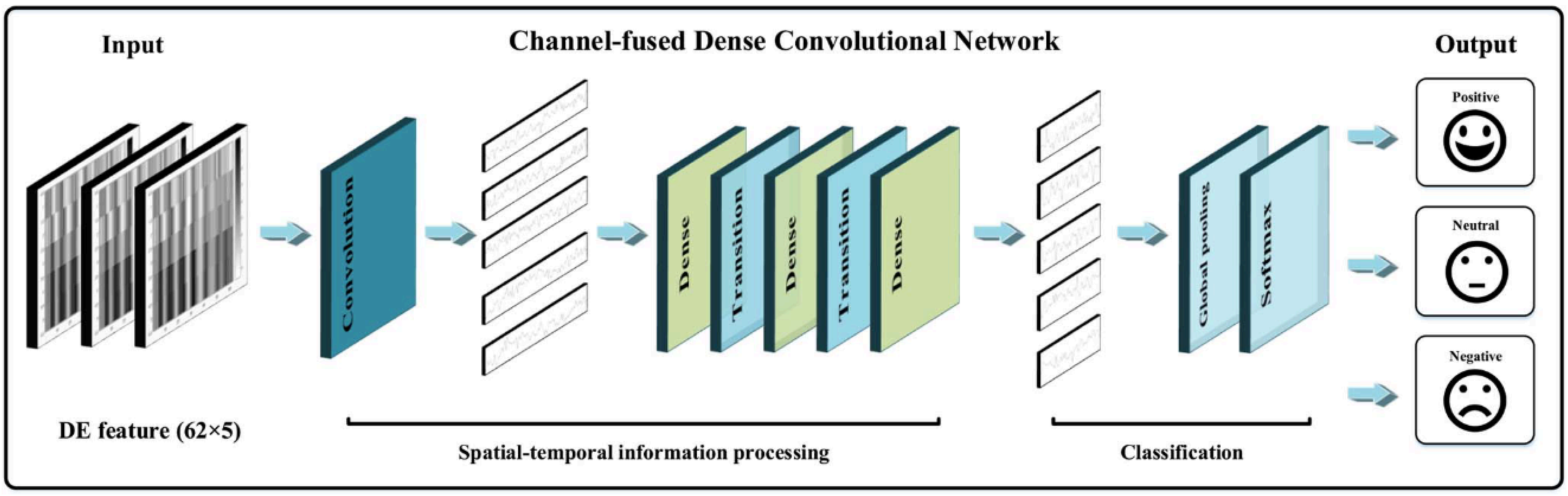


图 1. 基于脑电图的情绪识别所提出的方法的流程。

表 I
D D CDCN. T S C D THE O S C L . T S [] C K S , N F M

AND THE T L , R

Layers	Output size	CDCN (k=12)
Input	62 × 5	–
Convolution	62 × 1	[1×5, map 24, stride 1, conv]
Dense block 1	62 × 1	[3×1, map 36: 96, conv] ×6
Transition block 1	62 × 1	[1×1, map 96, conv]
	31 × 1	[2×1, stride 2, maxpooling]
Dense block 2	31 × 1	[3×1, map 108:168, conv] ×6
Transition block 2	31 × 1	[1×1, map 168, conv]
	16 × 1	[2×1, stride 2, maxpooling]
Dense block 3	16 × 1	[3×1, map 180:240, conv] ×6
Classification	240	global average pooling
	3	softmax

其中 [x, x, . . . , x] 表示 (l − 1) 层特征图的级联操作。

过渡块用于减小输入特征图的大小，这需要对它进行一些修改才能用于脑电图（EEG）信号。在实际应用中，过渡块由一个批量归一化层、一个卷积层和一个最大池化层组成。

在我们的实验中，CDCN 的深度被选为 24 层，具体细节见表 I。这里的输入维度和情绪数量对应于 SEED 数据集。增长率 k 和密集块的数量分别设置为 12 和 3。每个密集块包含六个卷积块，这些卷积块依次包含一个批量归一化[64]、一个修正线性激活（ReLU）[65]和一个 3×1 卷积，使用零填充以保持特征图的大小不变。在这种情况下，过渡块由一个 1×1 卷积和一个步长为 2 的 2×1 最大池化组成。

在该框架中，1×F 卷积层最初作为输入层执行，其特征图数量是增长率 k 的两倍，其中 N 表示每个电极的特征数量。三个密集块被连接在一起。同时，两个过渡块插入在两个相邻的密集块之间。

采用全局平均池化，随后接上最后一个密集块，然后附加一个 Softmax 分类器。在三个密集块中，特征图的大小分别为 62×1、31×1 和 16×1。如果每个密集块中的卷积产生 k 个特征图，则第 l 层有 k+k×l 个输出特征图，其中 k 表示密集块之前该层的特征图数量。对于每个过渡块中的卷积层，特征图的数量与前一卷积层相同。

C. 实现细节

使用反向传播通过时间（BPTT）算法优化网络参数，直到获得最优解或达到最大 epoch 数。此外，交叉熵目标函数被用作模型优化的损失函数[66]，其表达式为

$$Loss = cross\_entropy(y, y)$$

(4) 其中 y 和 y 分别表示训练样本的真实值和预测值。在训练过程中，目标函数 Loss 旨在降低损失值以提高模型的预测精度。我们使用 Adam 算法 [67] 以 10 的学习率训练 CDCN 框架。批处理大小设置为 64。验证集的真实值采用 one-hot 编码。模型基于 Keras 库实现，该库扩展自 Google Tensorflow。然后，我们通过监控验证集来保存最佳模型，使用 GeForce Titan X 进行模型训练大约需要二十分钟。

IV. E 在本节中，我们首先介绍 SEED 数据集，并提供关于数据集预处理的详细说明，该数据集用于评估所提出方法的性能。然后，我们分析模型在 SEED 数据集上的性能，并展示与其他先前用于类似研究的竞争性方法的性能比较。最后，

1 https://keras.io/ 2
https://www.tensorflow.org/

TABLE II
PERFORMANCE ON DIFFERENT FREQUENCY BANDS

Frequency band	Delta	Theta	Alpha	Beta	Gamma	All (%)
CDCN	65.19/6.83	69.84/8.37	72.16/8.49	80.83/7.64	82.63/8.01	**90.63/4.34**

we conduct experiments to evaluate the performance of the developed CDCN model on the DEAP dataset with method comparisons.

A. SEED Dataset and Preprocessing

The SEED dataset,[3] contributed by Duan *et al.* [31] and Zheng and Lu [68], focuses on EEG-based emotion recognition tasks. Fifteen emotion-evoking film clips with audios and scenes chosen as stimulus materials from the SEED dataset, could help collect high-quality signals. Three categories of emotions (positive, neutral and negative) were considered in the experiments. Each of the clips lasted for about four minutes and each emotion corresponds to five film clips.

Fifteen volunteering undergrads were selected to perform experiments with evaluation pasted on the Eysenck personality questionnaire (EPQ), which could assess one's personality traits [69]. Before each experiment, the subjects were advised to follow the procedure and refrained from unnecessary body movements. Each scalp EEG signal was collected by a 62-channel recording cap (ESI Neuroscan) and downsampled to the 200-Hz sampling rate. All electrodes were arranged in accordance with the standard 10–20 system [22]. The EEG signals were processed with a band-pass filter of 0.3–50 Hz to remove physiological and power frequency noises. Besides, face videos were recorded by a frontal camera. In each trial, the orders of executions were a 5 s tip, a film clip, a 45 s self-evaluation, and a 15 s rest. Each subject completed 15 trials for three sessions with an interval of one week or more between two sessions.

For each subject, the time durations are equal and fixed. We extract the EEG samples according to the duration of each movie, and then divide each channel of the EEG signal into the same-length segments of 1s without overlapping. The numbers of samples in three categories remain 1120, 1054, and 1070, respectively. Then, we compute the DE features on each EEG sample. One-hot state codes the sample representations for three categories. We take the same experimental protocol in [62], [68], and [70] to evaluate the performance of the CDCN model on the SEED dataset. For all the 15 trials from each session, the first nine trials are taken as the training set and the remaining six trials are taken as the testing set. Note that, the training data and the testing data are from different trials of the same session. The final recognition accuracy is averaged with the recognition accuracy of all the 15 subjects.

B. Overall Performance

The task can be regarded as a three-classification problem and CDCN is trained for EEG-based emotion recognition by

[3]http://bcmi.sjtu.edu.cn/home/seed/

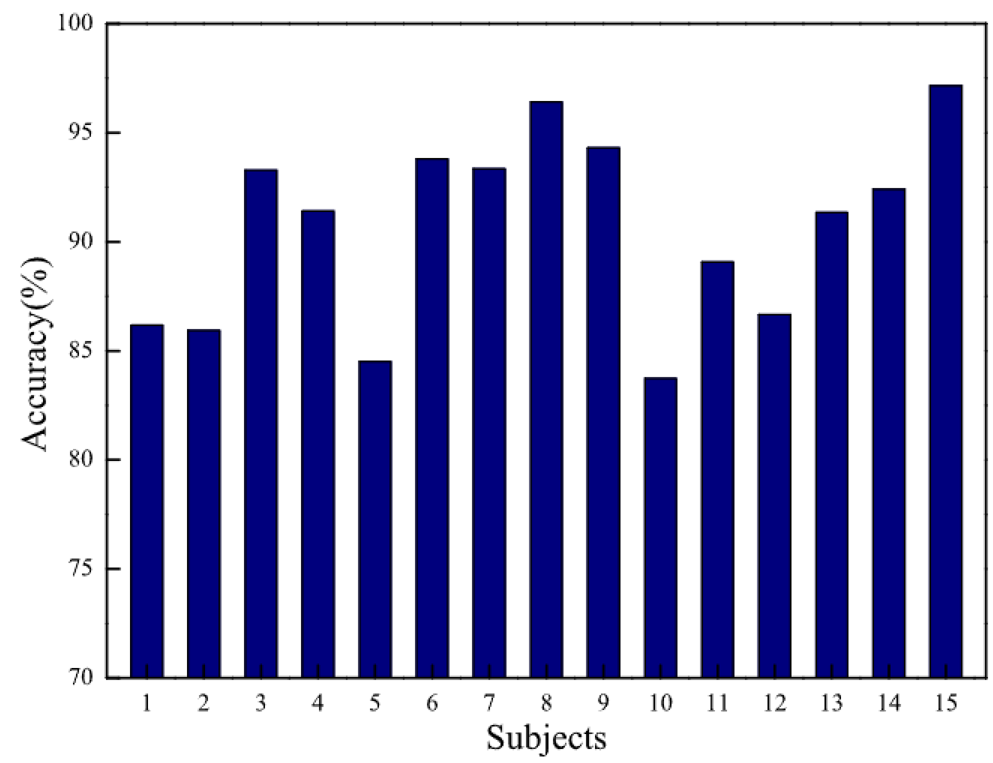


Fig. 2. Performances of the CDCN framework using the SEED dataset.

the dataset. In our experiments, the input sample is a 2-D feature matrix extracted from a 1s segment and the label prediction is exported for model evaluation. Besides, the individual performance is obtained by averaging the effects of three sessions. Fig. 2 presents the recognition accuracies of the CDCN framework for each subject using the SEED dataset.

From Fig. 2, we find that the CDCN algorithm for within subject is stably effective on the SEED dataset and all surpass 84%. The mean classification accuracy achieves 90.63% with a standard deviation 4.34%. The performance of eight subjects are over the mean accuracy, while seven are below it. Definite individual variations indeed exist, possibly caused by experimental situations and the subjects' physical conditions.

To reveal the relationship between different frequency bands and emotion states, the performance of the CDCN framework on different frequency bands (delta: 1–3 Hz, theta: 4–7 Hz, alpha: 8–13 Hz, beta: 14–30 Hz, and gamma: 31–50 Hz) are calculated using the SEED dataset, which is shown in Table II. Note that *All* in Table II denotes the performance uses the features of all the five frequency bands.

As observed in Table II, the accuracies of Beta and Gamma bands achieved 80% while the accuracies of three frequency bands (Delta, Theta, and Alpha) were lower than 80%. These results suggest that Beta and Gamma bands of brain activity are more related to emotional states than other frequency bands. The performance on All column works much better than these on five separate frequency bands. It shows that the features from all the five frequency bands contribute to the performance of the CDCN model on the SEED dataset, which indicates that the information related to emotion recognition tasks is distributed in different frequency bands of EEG signals.

表 II

P D F B

Frequency band	Delta	Theta	Alpha	Beta	Gamma	All (%)
CDCN	65.19/6.83	69.84/8.37	72.16/8.49	80.83/7.64	82.63/8.01	**90.63/4.34**

我们进行实验，以比较方法的方式评估所开发的 CDCN 模型在 DEAP 数据集上的性能。

A. SEED 数据集和预处理

SEED 数据集由 Duan 等人[31]和 Zheng 与 Lu[68]贡献，专注于基于脑电图的情绪识别任务。从 SEED 数据集中选取了 15 段带有音频和场景的情绪诱发电影片段作为刺激材料，有助于收集高质量的信号。实验中考虑了三种情绪类别（积极、中性和消极）。每段片段持续约四分钟，每种情绪对应五个电影片段。

十五名志愿者本科生被选中进行实验，并在艾森克人格问卷（EPQ）上进行了评估，该问卷可以评估一个人的性格特征[69]。在每次实验前，受试者被告知要遵循程序并避免不必要的身体运动。每个头皮脑电图信号由一个 62 通道的记录帽（ESI Neuroscan）收集，并降采样到 200 赫兹的采样率。所有电极都按照标准 10-20 系统[22]排列。脑电图信号通过 0.3-50 赫兹的带通滤波器进行处理，以去除生理和工频噪声。此外，还用前置摄像头记录了面部视频。在每个试验中，执行顺序为 5 秒的提示、视频片段、45 秒的自我评估和 15 秒的休息。每个受试者完成了三个会话，每个会话 15 次试验，两个会话之间间隔一周或更长时间。

对于每个受试者，时间间隔相等且固定。我们根据每部电影的时长提取脑电图样本，然后将脑电图信号的每个通道划分为长度相同的 1 秒片段，且不重叠。三类样本的数量分别保持为 1120、1054 和 1070。接着，我们在每个脑电图样本上计算 DE 特征。独热状态编码三类样本的表示。我们采用[62]、[68]和[70]中的相同实验方案，在 SEED 数据集上评估 CDCN 模型的性能。对于每个会话中的所有 15 次试验，前九次试验作为训练集，剩余六次试验作为测试集。请注意，训练数据和测试数据来自同一会话的不同试验。最终识别准确率是所有 15 个受试者识别准确率的平均值。

B. 整体性能

该任务可以被视为一个三分类问题，CDCN 通过脑电图进行情绪识别的训练。

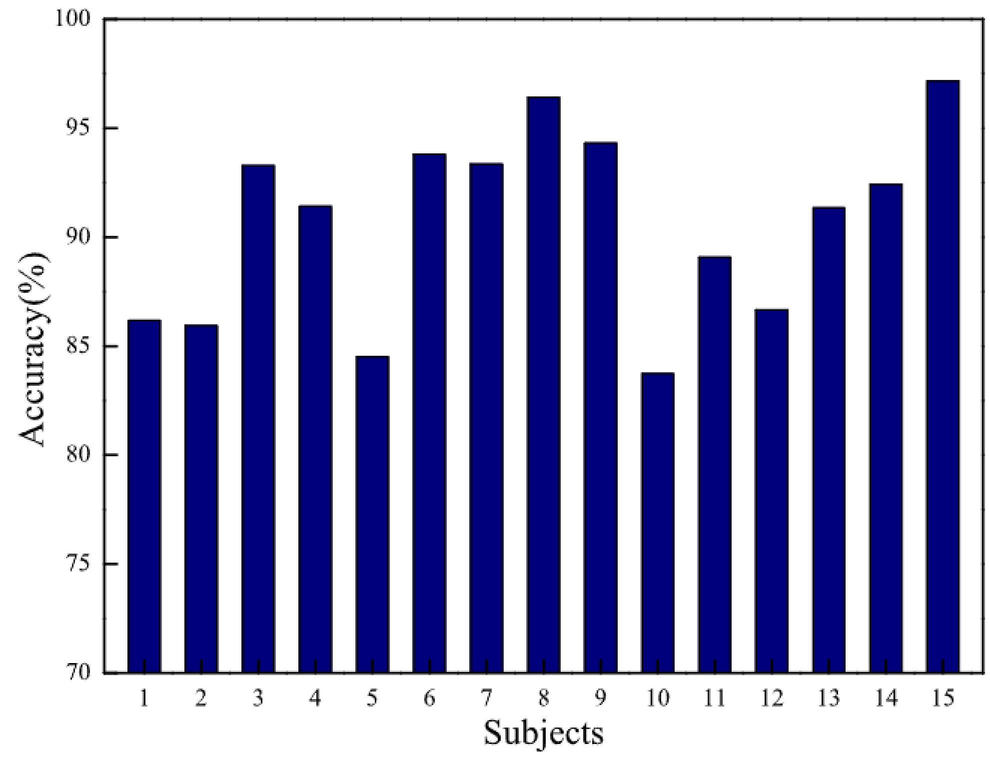


图 2. 使用 SEED 数据集的 CDCN 框架性能。

该数据集。在我们的实验中，输入样本是从 1 秒片段中提取的 2-D 特征矩阵，标签预测用于模型评估。此外，通过平均三个会话的效果来获得个体性能。图 2 展示了使用 SEED 数据集时，CDCN 框架对每个受试者的识别准确率。

从图 2 中，我们发现针对同一受试者的 CDCN 算法在 SEED 数据集上稳定有效，所有准确率均超过 84%。平均分类准确率达到 90.63%，标准差为 4.34%。八个受试者的性能超过平均准确率，而七个受试者低于平均准确率。确实存在显著的个体差异，这可能是由于实验环境和受试者的身体状况造成的。

为了揭示不同频段与情绪状态之间的关系，使用 SEED 数据集计算了 CDCN 框架在不同频段（δ：1–3 Hz，θ：4–7 Hz，α：8–13 Hz，β：14–30 Hz，以及γ：31–50 Hz）上的性能，如表 II 所示。请注意，表 II 中的"All"表示性能使用了所有五个频段的特征。

如表 II 所示，Beta 和 Gamma 频段的准确率达到了 80%，而 Delta、Theta 和 Alpha 三个频段的准确率均低于 80%。这些结果表明，脑活动的 Beta 和 Gamma 频段与情绪状态的关系比其他频段更为密切。在"All"列上的性能明显优于五个单独频段上的性能。这表明来自所有五个频段的特征共同促进了 CDCN 模型在 SEED 数据集上的性能，这表明与情绪识别任务相关的信息分布在脑电信号的各个频段中。

3 http://bcmi.sjtu.edu.cn/home/seed/

TABLE III
PERFORMANCE COMPARISONS ON THE SEED DATASET

Method	Description	Data type	Subject number	Accuracy (%)
SyncNet [71]	Convolutional neural network with Gaussian Process adapter	EEG	15	77.9
GSCCA [72]	Group sparse canonical correlation analysis with frequency features	EEG	15	83.72
DBN [68]	Deep belief network with DE features	EEG	15	86.08
HCNN [61]	Hierarchical convolutional neural network with DE features	EEG	4	86.2
SVM [68]	Support vector machine with DE features from 12 channels	EEG	15	86.65
MNN [73]	Minimalist neural network with reinforced gradient coefficients from 12 channels	EEG	15	88.23
STRNN [62]	Two-layer recurrent neural network with DE features	EEG	15	89.50
BDAE [70]	Bimodal deep autoencoder with DE features	EEG + EOG	9	91.01
GELM [54]	Graph regularized extreme learning machine with DE features	EEG	15	91.07
CDCN	Channel-fused dense convolutional network with DE features	EEG	15	**90.63**

The confusion matrix of the CDCN framework evaluated on the SEED dataset is presented in Fig. 3, which shows the classification accuracy of each emotion. The value (i, j) denotes the percentage of samples in class i that is classified as class j. As shown in Fig. 3, our method shows a good performance in recognizing all three types of emotions and the accuracies of them exceed 82%. Positive emotion is recognized with higher accuracies, while negative emotion is relatively difficult to recognize as it is partly confused with the neutral.

These results reflect that our CDCN model provides an effective relationship between emotional states and EEG signals. But, from another perspective, we suggest taking the factors of interclass variations into consideration to establish a more robust emotion recognition framework, which could be our future research work.

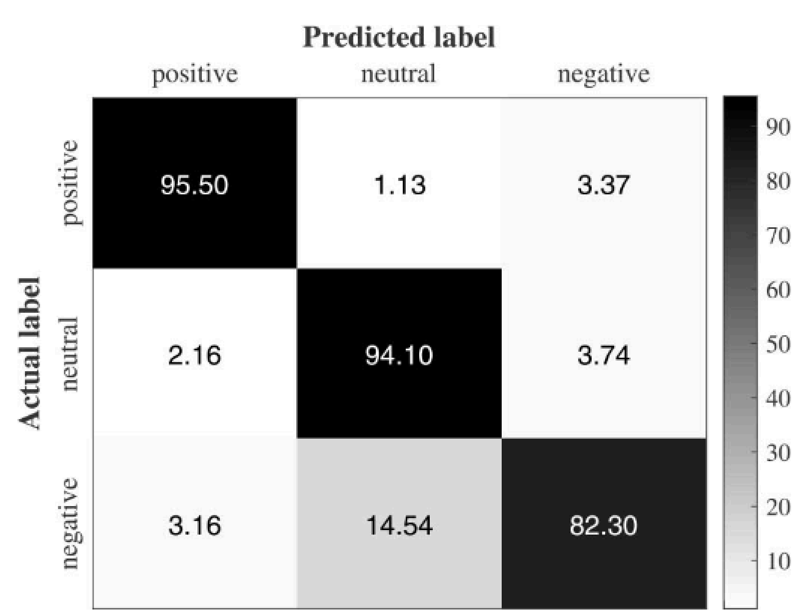


Fig. 3. Confusion matrix of the CDCN framework evaluated on the SEED dataset.

C. Comparison With Previous Studies

In recent years, there are growing interests in the publicly available dataset SEED, based on which various studies have been conducted to explore the challenging task of emotion recognition. Despite some differences in detailed treatments, the existing approaches provide valuable research ideas and findings. Here, we select some competitive studies focusing on the SEED dataset for comparisons.

Table III shows the performances and other details of the following methods: SyncNet [71], GSCCA [72], DBN [68], HCNN [61], SVM [68], MNN [73], STRNN [62], BDAE [70], and GELM [54]. Most of these methods employed all the subjects from the SEED dataset and take their EEG signals for analysis, while HCNN and BDAE considered four and nine subjects, respectively. Note that BDAE gets additional terms from eye movement combined with EEG signals. Therefore, we also took these differences into consideration in the performance comparisons.

DE features are employed in a large part of the above studies, except SyncNet. SyncNet gets the mean accuracy of 77.9% but it performs worse than other DE-related methods, which reflects the crucial correlation between DE features and emotional states. The performance of these methods combined with DE features changes between 83.72% and 91.07%, while GELM achieves a higher accuracy of 91.07%. Moreover, various neural network frameworks are used to recognize different emotional states, except GSCCA and SVM. STRNN with accuracy of 89.50% benefits a lot from the spatial–temporal structure, while BDAE and GELM methods also manage DE features well, and both have performances above 91%. Note that, the effects of BDAE and GELM are slightly better than our framework, with higher standard deviations of 8.91% and 7.54%, respectively. Besides, BDAE employs the recorded data of 9 subjects and additional eye movement signals.

Among these ten methods, CDCN shows an excellent performance for emotion recognition tasks through EEG signals with considerable enhancement compared with other

表 III
P C OMPARISONS ON THE SEED D

Method	Description	Data type	Subject number	Accuracy (%)
SyncNet [71]	Convolutional neural network with Gaussian Process adapter	EEG	15	77.9
GSCCA [72]	Group sparse canonical correlation analysis with frequency features	EEG	15	83.72
DBN [68]	Deep belief network with DE features	EEG	15	86.08
HCNN [61]	Hierarchical convolutional neural network with DE features	EEG	4	86.2
SVM [68]	Support vector machine with DE features from 12 channels	EEG	15	86.65
MNN [73]	Minimalist neural network with reinforced gradient coefficients from 12 channels	EEG	15	88.23
STRNN [62]	Two-layer recurrent neural network with DE features	EEG	15	89.50
BDAE [70]	Bimodal deep autoencoder with DE features	EEG + EOG	9	91.01
GELM [54]	Graph regularized extreme learning machine with DE features	EEG	15	91.07
CDCN	Channel-fused dense convolutional network with DE features	EEG	15	**90.63**

在 SEED 数据集上评估的 CDCN 框架的混淆矩阵如图 3 所示，该图显示了每种情绪的分类准确率。值(i, j)表示属于类别 i 的样本中被分类为类别 j 的百分比。如图 3 所示，我们的方法在识别三种类型的情绪方面表现出良好性能，它们的准确率均超过 82%。积极情绪的识别准确率较高，而消极情绪的识别相对较难，因为它部分与中性情绪混淆。

这些结果反映出我们的 CDCN 模型在情绪状态与脑电图信号之间建立了有效的关系。然而，从另一个角度来看，我们建议考虑类间差异因素，以建立更鲁棒的情绪识别框架，这可以成为我们未来的研究工作。

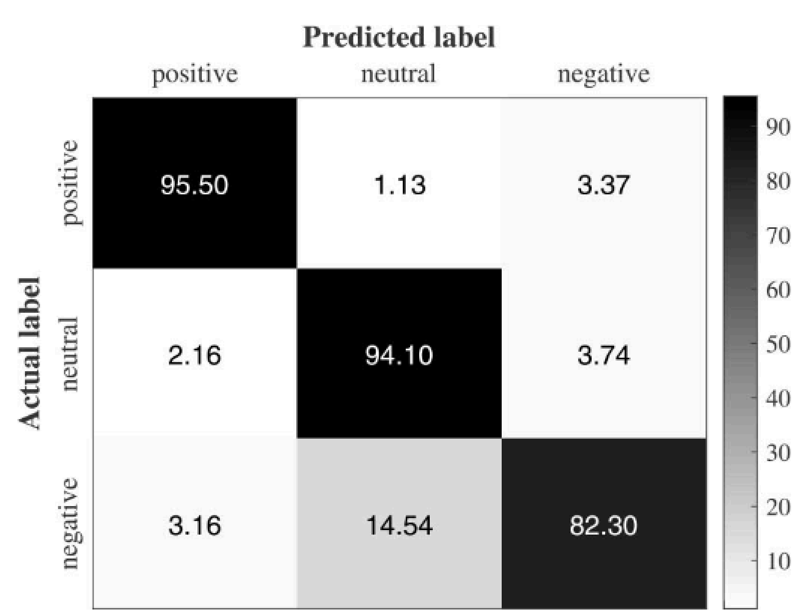


图 3. 在 SEED 数据集上评估的 CDCN 框架的混淆矩阵。

C. 与先前研究的比较

近年来，公开数据集 SEED 引起了越来越多的兴趣，基于该数据集进行了各种研究，以探索具有挑战性的情绪识别任务。尽管在详细处理上存在一些差异，但现有方法提供了宝贵的研究思路和发现。在这里，我们选择了一些针对 SEED 数据集的竞争性研究进行比较。

表 III 展示了以下方法的性能和其他详细信息：SyncNet [71]、GSCCA [72]、DBN [68]、HCNN [61]、SVM [68]、MNN [73]、STRNN [62]、BDAE [70]和 GELM [54]。这些方法中大多数使用了 SEED 数据集的所有受试者及其脑电信号进行分析，而 HCNN 和 BDAE 分别考虑了四个和九个受试者。请注意，BDAE 通过结合眼动和脑电信号获得了额外的信息。因此，在性能比较中，我们也考虑了这些差异。

DE 特征被上述大部分研究采用，SyncNet 除外。SyncNet 取得了 77.9%的平均准确率，但表现不如其他与 DE 相关的方法，这反映了 DE 特征与情绪状态之间的关键关联。结合 DE 特征的方法性能在 83.72%到 91.07%之间变化，而 GELM 达到了更高的 91.07%准确率。此外，除了 GSCCA 和 SVM 外，各种神经网络框架被用于识别不同的情绪状态。STRNN 以 89.50%的准确率显著受益于时空结构，而 BDAE 和 GELM 方法也能很好地处理 DE 特征，两者性能均超过 91%。值得注意的是，BDAE 和 GELM 的效果略优于我们的框架，其标准差分别为 8.91%和 7.54%。此外，BDAE 采用了 9 名受试者的记录数据以及额外的眼动信号。

在这十种方法中，CDCN 通过脑电信号进行情绪识别任务表现出色，与其他方法相比有显著提升。

TABLE IV
PERFORMANCE COMPARISONS ON THE DEAP DATASET

Study	Description	Accuracy (%)	
		Valence	Arousal
RVM [74]	Relevance vector machine with graph-theoretic features	69	67
C-RNN [75]	Convolutional recurrent neural network with wavelet transform-based features	72.06	74.12
CNN [76]	Convolutional neural network with 101 extracted features per channel	81.41	73.36
MDL [70]	SVM with the pre-trained features from the bimodal deep autoencoder	85.2	80.5
WT-SVM [77]	SVM with the extracted features from discrete wavelet transform method	84.95	84.14
DBN [78]	Deep belief network with power spectral density features	88.33	88.59
CDCN	Channel-fused dense convolutional network with DE features	**92.24**	**92.92**

methods. This indicates that our CDCN framework can robustly capture valid information from EEG signals owing to its good handling of electrode correlations and temporal dependencies.

D. Experiments on DEAP Dataset

In this part, we conduct experiments to evaluate the performance of the developed CDCN model on the DEAP[4] dataset. The DEAP dataset contained EEG and peripheral physiological signals of 32 subjects (50 percent females), which were collected by watching 40 one-minute music videos. The subjects were asked to perform self-assessment of arousal, valence, liking, and dominance on a score from 1 to 9 for each video. The EEG signals were recorded using 32 active AgCl electrodes and downsampled to the 128-Hz sampling rate. Then, a band-pass filter from 4 Hz to 45 Hz was applied. These two preprocessing steps were pregiven in the DEAP dataset [13]. The data is segmented into 1 s samples without overlapping. To compare the performance with previous studies, we construct two classification tasks based on the VA model: low/high valence (task1) and low/high arousal (task2). Besides, for the labels of these two tasks, the self-assessment score between 1 and 4.8 is low and the value between 5.2 and 9 is high. We take the same experimental protocol in [70] to evaluate the performance of the CDCN model on the DEAP dataset. For all the 40 trials of each subject, the samples from 36 trials are taken as the training set and the samples from the remaining 4 trials are taken as the testing set, which can avoid dependency between the training and testing set. The final recognition accuracy is averaged with the recognition accuracy of all the 32 subjects.

Table IV shows the experimental results of the CDCN method with respect to the emotion dimensions (including valence and arousal). We compare the CDCN results with the effects of six existing studies on the DEAP dataset, including RVM [74], C-RNN [75], CNN [76], MDL [70], WT-SVM [77], and DBN [78]. These existing studies used different extracted features and different classifiers. Note that GELM [54] was also evaluated on the DEAP dataset, but using a subject-independent validation scheme with sample shuffling. Therefore, we do not include GELM [54] in the method comparisons on the DEAP dataset.

From Table IV, the average accuracies of the developed CDCN model are 92.24% and 92.92% for two different tasks, respectively. The performance of six compared methods changes between 67% and 89%. Results show that the CDCN model performs much better than the other six compared methods on the DEAP dataset. Moreover, the trained model can be efficiently applied in online BCI recognition tasks.

V. DISCUSSION

The above comparison analysis indicates that the CDCN framework performs better than most of the existing methods, which combines prior knowledge with the developed model to co-train this task. This can be primarily attributed to the 1-D convolution along the temporal dimension in the CDCN framework. It provides an interpretable sense for efficient weighted combinations of contextual features using trained filters. For example, a $1 \times F$ convolution could model the input sample of size $[E, F]$, with E electrodes and F features. Several feature maps of size $[E, 1]$ are obtained after proper training, which denotes adaptive contextual features. There is one component for handling the temporal information, which performs well among the compared methods with different DL frameworks.

Another observation is the employment of 1-D dense block. Dense block encourages feature reuse across the whole network and is sufficient in feature maps with low dimensions. It gains electrode correlations along the spatial dimension through EEG signals. Notably, our CDCN receives excellent performance compared with other DL-based methods, which emphasizes the importance of spatial–temporal information. Especially, the feature reuse of 1-D dense block can help investigate the information of electrode correlations faster, which is the key point of the CDCN model performs notably well among these compared methods. Furthermore, the design principles of the CDCN model can be employed by broader EEG-based recognition tasks. Existing studies have developed some novel methods to conduct channel selection [68], [79], [80]. In [79], the extracted features like synchronization likelihood were used to reduce the number of EEG channels, which led to a slight loss of the classification accuracy rate for emotion assessment. In [68], the critical channels were found through analyzing the weight distributions of the trained DBNs. Our future work will focus on using fewer critical channels to train the recognition networks.

VI. CONCLUSION

In this article, a novel CDCN model is proposed to provide robust representations for emotion recognition from EEG

[4]http://www.eecs.qmul.ac.uk/mmv/datasets/deap/

表 IV
P C OMPARISONS ON THE DEAP D

Study	Description	Accuracy (%)	
		Valence	Arousal
RVM [74]	Relevance vector machine with graph-theoretic features	69	67
C-RNN [75]	Convolutional recurrent neural network with wavelet transform-based features	72.06	74.12
CNN [76]	Convolutional neural network with 101 extracted features per channel	81.41	73.36
MDL [70]	SVM with the pre-trained features from the bimodal deep autoencoder	85.2	80.5
WT-SVM [77]	SVM with the extracted features from discrete wavelet transform method	84.95	84.14
DBN [78]	Deep belief network with power spectral density features	88.33	88.59
CDCN	Channel-fused dense convolutional network with DE features	**92.24**	**92.92**

方法。这表明我们的 CDCN 框架能够稳健地从脑电图信号中捕获有效信息，这得益于其对电极相关性和时间依赖性的良好处理。

D. DEAP 数据集上的实验

在本部分，我们通过实验评估所开发的 CDCN 模型在 DEAP 数据集上的性能。DEAP 数据集包含了 32 名受试者（其中 50%为女性）的脑电图（EEG）和外周生理信号，这些数据是在观看 40 个一分钟的音乐视频时收集的。受试者被要求对每个视频进行自我评估，从 1 到 9 的分数来评价唤醒度、效价、喜好度和支配度。脑电图信号使用 32 个活性 AgCl 电极记录，并降采样到 128 赫兹的采样率。然后，应用了 4 赫兹到 45 赫兹的带通滤波器。这两个预处理步骤在 DEAP 数据集[13]中是预先给定的。数据被分割成 1 秒的样本，且无重叠。为了与先前研究进行比较，我们基于 VA 模型构建了两个分类任务：低/高效价（任务 1）和低/高唤醒度（任务 2）。此外，对于这两个任务的标签，1 到 4.8 的分数为低，5.2 到 9 的分数为高。我们采用文献[70]中的相同实验方案来评估 CDCN 模型在 DEAP 数据集上的性能。 对于每个被试的 40 次试验，从其中的 36 次试验中选取样本作为训练集，从剩余的 4 次试验中选取样本作为测试集，这样可以避免训练集和测试集之间的依赖性。最终识别准确率通过对所有 32 个被试的识别准确率进行平均得到。

表 IV 展示了针对情绪维度（包括效价和唤醒度）的 CDCN 方法的实验结果。

我们将 CDCN 的结果与在 DEAP 数据集上的六项现有研究的效果进行了比较，包括 RVM [74]、C-RNN [75]、CNN [76]、MDL [70]、WT-SVM [77]和 DBN [78]。这些现有研究使用了不同的提取特征和不同的分类器。需要注意的是，GELM [54]也在 DEAP 数据集上进行了评估，但使用的是与被试无关的验证方案，并且对样本进行了随机打乱。因此，我们不将 GELM [54]包括在 DEAP 数据集的方法比较中。

从表 IV 可以看出，开发的 CDCN 模型在两个不同任务上的平均准确率分别为 92.24%和 92.92%。六个对比方法的性能在 67%到 89%之间变化。结果表明，CDCN 模型在 DEAP 数据集上表现远优于其他六个对比方法。此外，训练好的模型可以高效应用于在线 BCI 识别任务。

V. D 上述对比分析表明，CDCN 框架的性能优于大多数现有方法，它结合先验知识与开发模型共同训练该任务。这主要归因于 CDCN 框架中沿时间维度的一维卷积。它为使用训练好的滤波器高效组合上下文特征提供了可解释性。例如，一个 1×F 卷积可以建模大小为[E, F]的输入样本，其中 E 为电极数量，F 为特征数量。经过适当训练后，可以得到多个大小为[E, 1]的特征图，这表示自适应的上下文特征。有一个组件用于处理时间信息，在对比的不同深度学习框架方法中表现良好。

另一个观察是采用了 1-D 密集块。密集块鼓励在整个网络中重用特征，并且在低维特征图中足够有效。它通过脑电图信号在空间维度上获得电极相关性。值得注意的是，与其他基于深度学习的方法相比，我们的 CDCN 取得了优异的性能，这突出了时空信息的重要性。特别是，1-D 密集块的特征重用有助于更快地研究电极相关性的信息，这是 CDCN 模型在这些比较方法中表现突出的关键点。此外，CDCN 模型的设计原则可以应用于更广泛的基于脑电图的情绪识别任务。现有研究已经开发了一些新颖的方法来进行通道选择[68]、[79]、[80]。在[79]中，提取的特征如同步性指数被用于减少脑电图通道数量，这导致情绪评估的分类准确率略有下降。在[68]中，通过分析训练好的 DBN 的权重分布找到了关键通道。 我们的未来工作将集中于使用较少的关键通道来训练识别网络。

VI. C 在本文中，提出了一种新型 CDCN 模型，为脑电图（EEG）的情绪识别提供鲁棒表示

4 http://www.eecs.qmul.ac.uk/mmv/datasets/deap/

signals. Dense block is used to encourage feature reuse on image-based classification tasks, and the 1-D dense block is reformulated from the 2-D dense block so as to collect electrode correlations from EEG signals. On the other hand, inspired by the basic components of convolution, a 1-D convolution is employed to receive proper weighted combinations of contextual features to deal with temporal information. Consequently, with the above two advances, the proposed CDCN is used to target multichannel series, making it quite proper for managing the spatial–temporal information from EEG signals. The experimental results demonstrate that the CDCN model has excellent performances compared with other existing methods when evaluating on the SEED and DEAP datasets.

One possible improvement of our framework is to integrate signal sequence with prior knowledge. Although the CDCN model outperforms several DL methods with extracted features, prior knowledge could be combined with EEG sequences to further strengthen the model for even better performances with positive effects. In addition, 1-D convolution plays the role of EEG weighted feature combinations, which greatly contributes to the computational efficiency. Especially, if one examines the weights of the trained networks, more crucial information about the weighted feature combinations could be used for EEG feature selection. It is promising that fewer extracted features can gain better performance and deeply discover the contribution between task labels and the features in different frequency bands. The method has good task-adaptive ability and can be applied to other different EEG classification tasks, including fatigue recognition and motor imagery. Therefore, following-up studies could be conducted to further improve the performance on different EEG classification tasks. Looking forward, with excellent real-time performances, the CDCN model could be effectively employed to health monitoring tasks, and extended for broader BCI recognition tasks.

References

[1] R. Pandey and A. K. Choubey, "Emotion and health: An overview," *SIS J. Projective Psychol. Mental Health*, vol. 17, no. 2, pp. 135–152, 2010.

[2] S. D. Pressman and S. Cohen, "Does positive affect influence health?" *Psychol. Bull.*, vol. 131, no. 6, pp. 925–971, 2005.

[3] E. Mumford, H. J. Schlesinger, and G. V. Glass, "The effect of psychological intervention on recovery from surgery and heart attacks: An analysis of the literature," *Amer. J. Public Health*, vol. 72, no. 2, pp. 141–151, 1982.

[4] R. W. Picard, *Affective Computing*. Cambridge, MA, USA: MIT Press, 1997.

[5] R. A. Calvo and S. D'Mello, "Affect detection: An interdisciplinary review of models, methods, and their applications," *IEEE Trans. Affect. Comput.*, vol. 1, no. 1, pp. 18–37, Jan. 2010.

[6] Z. Zeng, M. Pantic, G. I. Roisman, and T. S. Huang, "A survey of affect recognition methods: Audio, visual, and spontaneous expressions," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 31, no. 1, pp. 39–58, Jan. 2009.

[7] R. Gravina, P. Alinia, H. Ghasemzadeh, and G. Fortino, "Multi-sensor fusion in body sensor networks: State-of-the-art and research challenges," *Inf. Fusion*, vol. 35, pp. 68–80, May 2017.

[8] E. Bekele, D. Bian, J. Peterman, S. Park, and N. Sarkar, "Design of a virtual reality system for affect analysis in facial expressions (VR-SAAFE); Application to Schizophrenia," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 25, no. 6, pp. 739–749, Jun. 2017.

[9] A. Mencattini, E. Martinelli, F. Ringeval, B. Schuller, and C. D. Natale, "Continuous estimation of emotions in speech by dynamic cooperative speaker models," *IEEE Trans. Affect. Comput.*, vol. 8, no. 3, pp. 314–327, Jul.–Sep. 2017.

[10] S. M. Mohammad and P. D. Turney, "Crowdsourcing a word-emotion association lexicon," *Comput. Intell.*, vol. 29, no. 3, pp. 436–465, Aug. 2013.

[11] F. Agrafioti, D. Hatzinakos, and A. K. Anderson, "ECG pattern analysis for emotion detection," *IEEE Trans. Affect. Comput.*, vol. 3, no. 1, pp. 102–115, Jan.–Mar. 2012.

[12] K. Tanaka, M. Tanaka, T. Kajiwara, and H. Q. Wang, "A practical SSVEP-based algorithm for perceptual dominance estimation in binocular rivalry," *IEEE Trans. Cogn. Develop. Syst.*, vol. 10, no. 2, pp. 476–482, Jun. 2018.

[13] S. Koelstra *et al.*, "DEAP: A database for emotion analysis; using physiological signals," *IEEE Trans. Affect. Comput.*, vol. 3, no. 1, pp. 18–31, Jan.–Mar. 2012.

[14] X.-W. Wang, D. Nie, and B.-L. Lu, "Emotional state classification from EEG data using machine learning approach," *Neurocomputing*, vol. 129, pp. 94–106, Apr. 2014.

[15] P. Buitelaar *et al.*, "MixedEmotions: An open-source toolbox for multimodal emotion analysis," *IEEE Trans. Multimedia*, vol. 20, no. 9, pp. 2454–2465, Sep. 2018.

[16] A. R. Damasio *et al.*, "Subcortical and cortical brain activity during the feeling of self-generated emotions," *Nat. Neurosci.*, vol. 3, no. 10, pp. 1049–1056, 2000.

[17] Y. M. Chi, Y. T. Wang, Y. J. Wang, C. Maier, T. P. Jung, and G. Cauwenberghs, "Dry and noncontact EEG sensors for mobile brain–computer interfaces," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 20, no. 2, pp. 228–235, Mar. 2012.

[18] Y. Tu, Y. S. Hung, L. Hu, G. Huang, Y. Hu, and Z. Zhang, "An automated and fast approach to detect single-trial visual evoked potentials with application to brain–computer interface," *Clin. Neurophysiol.*, vol. 125, no. 12, pp. 2372–2383, Mar. 2014.

[19] M. H. Silber *et al.*, "The visual scoring of sleep in adults," *J. Clin. Sleep Med.*, vol. 3, no. 2, pp. 121–131, 2007.

[20] J. W. Cao, J. H. Zhu, W. B. Hu, and A. Kummert, "Epileptic signal classification with deep EEG features by stacked CNNs," *IEEE Trans. Cogn. Develop. Syst.*, early access, doi: 10.1109/TCDS.2019.2936441.

[21] A. Sahayadhas, K. Sundaraj, and M. Murugappan, "Detecting driver drowsiness based on sensors: A review," *Sensors*, vol. 12, no. 12, pp. 16937–16953, Dec. 2012.

[22] G. H. Klem, H. O. Luders, H. Jasper, and C. Elger, "The ten-twenty electrode system of the international federation," *Electroencephalography Clin. Neurophysiol.*, vol. 52, no. 3, pp. 3–6, 1999.

[23] V. Vanitha and P. Krishnan, "Time-frequency analysis of EEG for improved classification of emotion," *Int. J. Biomed. Eng. Technol.*, vol. 23, nos. 2–4, pp. 191–212, 2017.

[24] Z. K. Gao, Q. Cai, Y. X. Yang, N. Dong, and S. S. Zhang, "Visibility graph from adaptive optimal kernel time-frequency representation for classification of epileptiform EEG," *Int. J. Neural. Syst.*, vol. 27, no. 4, Jun. 2017, Art. no. 1750005.

[25] Z. K. Gao *et al.*, "An adaptive optimal-kernel time-frequency representation-based complex network method for characterizing fatigued behavior using the SSVEP-based BCI system," *Knowl. Based. Syst.*, vol. 152, pp. 163–171, Jul. 2018.

[26] L. Cheng *et al.*, "Principal states of dynamic functional connectivity reveal the link between resting-state and task-state brain: An fMRI study," *Int. J. Neural Syst.*, vol. 28, no. 7, Apr. 2018, Art. no. 1850002.

[27] H. Zhu *et al.*, "Abnormal dynamic functional connectivity associated with subcortical networks in Parkinsons disease: A temporal variability perspective," *Front. Neurosci.*, vol. 13, p. 80, Feb. 2019, doi: 10.3389/fnins.2019.00080.

[28] D. Garrett, D. A. Peterson, C. W. Anderson, and M. H. Thaut, "Comparison of linear, nonlinear, and feature selection methods for EEG signal classification," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 11, no. 2, pp. 141–144, Jun. 2003.

[29] Z. K. Gao, W. D. Dang, M. X. Liu, W. Guo, K. Ma, and G. R. Chen, "Classification of EEG signals on VEP-based BCI systems with broad learning," *IEEE Trans. Syst., Man, Cybern., Syst.*, early access, doi: 10.1109/TSMC.2020.2964684.

[30] G. L. Ahern and G. E. Schwartz, "Differential lateralization for positive and negative emotion in the human brain: EEG spectral analysis," *Neuropsychologia*, vol. 23, no. 6, pp. 745–755, 1985.

[31] R. N. Duan, J. Y. Zhu, and B. L. Lu, "Differential entropy feature for EEG-based emotion classification," in *Proc. IEEE/EMBS Conf. Neur. Eng.*, San Diego, CA, USA, 2013, pp. 81–84.

信号。在基于图像的分类任务中，密集块用于鼓励特征重用，而 1-D 密集块是从 2-D 密集块重新构建的，以便从脑电图信号中收集电极相关性。另一方面，受卷积基本组件的启发，采用 1-D 卷积来接收适当的上下文特征组合，以处理时间信息。因此，通过上述两个进步，所提出的 CDCN 被用于针对多通道序列，非常适合管理脑电图信号的空间-时间信息。实验结果表明，与其他现有方法相比，在 SEED 和 DEAP 数据集上评估时，CDCN 模型表现出色。

我们框架的一个可能改进是将信号序列与先验知识相结合。尽管 CDCN 模型在提取特征方面优于多种深度学习方法，但先验知识可以与脑电图序列结合，进一步增强模型，以获得更好的性能和积极效果。此外，一维卷积在脑电图加权特征组合中发挥作用，极大地提高了计算效率。特别是，如果检查训练网络的权重，则可以更多地利用关于加权特征组合的关键信息来进行脑电图特征选择。更少提取的特征可以获得更好的性能，并深入发现任务标签与不同频段特征之间的贡献。该方法具有良好的任务自适应能力，可以应用于其他不同的脑电图分类任务，包括疲劳识别和运动想象。因此，后续研究可以进一步改进在不同脑电图分类任务上的性能。 展望未来，凭借出色的实时性能，CDCN 模型可以有效地应用于健康监测任务，并扩展至更广泛的 BCI 识别任务。

R

[1] R. Pandey 和 A. K. Choubey, “情绪与健康：概述,” SIS J. 项目心理学与心理健康, 卷 17, 第 2 期, pp. 135–152, 2010. [2] S. D. Pressman 和 S. Cohen, “积极情绪是否影响健康?” 心理学公报, 卷 131, 第 6 期, pp. 925–971, 2005. [3] E. Mumford, H. J. Schlesinger, 和 G. V. Glass, “心理干预对手术和心脏病康复的影响：文献分析,” 美国公共卫生杂志, 卷 72, 第 2 期, pp. 141–151, 1982. [4] R. W. Picard,《情感计算》. 马萨诸塞州剑桥, USA: MIT 出版社,

1997. [5] R. A. Calvo 和 S. D’Mello, “情感检测：一个跨学科

“模型、方法和它们的应用的综述”，IEEE 情感计算汇刊，第 1 卷，第 1 期，第 18-37 页，2010 年 1 月。

[6] Z. Zeng, M. Pantic, G. I. Roisman, 和 T. S. Huang, “情感识别方法综述：音频、视觉和自发表达,” IEEE 模式分析及机器智能汇刊, 第 31 卷, 第 1 期, 第 39–58 页, 2009 年 1 月. [7] R. Gravina, P. Alinia, H. Ghasemzadeh, 和 G. Fortino, “体感网络中的多传感器融合：现状和研究挑战,” 信息融合汇刊, 第 35 卷, 第 68–80 页, 2017 年 5 月. [8] E. Bekele, D. Bian, J. Peterman, S. Park, 和 N. Sarkar, “用于面部表情情感分析的虚拟现实系统设计 (VR-SAAFE)；应用于精神分裂症,” IEEE 神经系统及康复工程汇刊, 第 25 卷, 第 6 期, 第 739–749 页, 2017 年 6 月.

[9] A. Mencattini, E. Martinelli, F. Ringeval, B. Schuller, 和 C. D. Natale, “通过动态协同说话人模型对语音中情感的持续估计,” IEEE 情感计算汇刊, 第 8 卷, 第 3 期, 第 314–327 页, 2017 年 7 月–9 月.

[10] S. M. Mohammad 和 P. D. Turney, “众包构建词-情绪关联词典,” 计算机智能, 第 29 卷, 第 3 期, pp. 436–465, 2013 年 8 月. [11] F. Agrafioti, D. Hatzinakos, 和 A. K. Anderson, “心电图模式分析用于情绪检测,” IEEE 情感计算汇刊, 第 3 卷, 第 1 期, pp. 102–115, 2012 年 1 月–3 月. [12] K. Tanaka, M. Tanaka, T. Kajiwara, 和 H. Q. Wang, “一种基于 SSVEP 的双目竞争感知优势估计实用算法,” IEEE 认知发展系统汇刊, 第 10 卷, 第 2 期, pp. 476–482, 2018 年 6 月. [13] S. Koelstra 等, “DEAP: 用于情绪分析的数据库；使用生理信号,” IEEE 情感计算汇刊, 第 3 卷, 第 1 期, pp. 18–31, 2012 年 1 月–3 月. [14] X.-W. Wang, D. Nie, 和 B.-L. Lu, “基于机器学习方法从脑电数据分类情绪状态,” 神经计算, 第 129 期, pp. 94–106, 2014 年 4 月. [15] P. Buitelaar 等, “MixedEmotions: 用于多模态情绪分析的开放源代码工具箱,” IEEE 多媒体汇刊, 第 20 卷, 第 9 期, pp. 2454–2465, 2018 年 9 月. [16] A. R. Damasio 等, “自我生成情绪时的脑皮层和皮层下脑活动,” 自然神经科学, 第 3 卷, 第 10 期, pp. 1049–1056, 2000. [17] Y. M. Chi, Y. T. Wang, Y. J. Wang, C. Maier, T. P. Jung, 和 G. Cauwenberghs, “用于移动脑机接口的干式和非接触式脑电图传感器,” IEEE 交易. 神经系统康复工程, 卷. 20, 第. 2, 页. 228–235, 2012 年 3 月. [18] Y. Tu, Y. S. Hung, L. Hu, G. Huang, Y. Hu, 和 Z. Zhang, “一种用于脑机接口的单次视觉诱发电位自动快速检测方法,” 临床神经生理学, 卷. 125, 第. 12, 页. 2372–2383, 2014 年 3 月. [19] M. H. Silber 等, “成年人睡眠的视觉评分,” 临床睡眠医学杂志, 卷. 3, 第. 2, 页. 121–131, 2007. [20] J. W. Cao, J. H. Zhu, W. B. Hu, 和 A. Kummert, “基于深度脑电图特征的堆叠卷积神经网络进行癫痫信号分类,” IEEE 交易. 认知发展系统, 早期访问, doi: 10.1109/TCDS.2019.2936441. [21] A. Sahayadhas, K. Sundaraj, 和 M. Murugappan, “基于传感器检测驾驶员困倦: 一篇综述,” 传感器, 卷. 12, 第. 12, 页. 16937–16953, 2012 年 12 月. [22] G. H. Klem, H. O. Luders, H. Jasper, 和 C. Elger, “国际联合会二十电极系统,” 脑电图与临床神经生理学, 卷. 52, 第. 3, 页. 3–6, 1999. [23] V. Vanitha 和 P. Krishnan, “脑电图的时间-频率分析以改进情绪分类,” 国际生物医学工程与技术杂志, 第 23 卷, 第 2-4 期, 第 191-212 页, 2017. [24] Z. K. Gao, Q. Cai, Y. X. Yang, N. Dong 和 S. S. Zhang, “自适应最优核时间-频率表示的可见图用于癫痫脑电图分类,” 国际神经系统杂志, 第 27 卷, 第 4 期, 2017 年 6 月, 第 1750005 号文章. [25] Z. K. Gao 等人, “基于自适应最优核时间-频率表示的复杂网络方法用于使用基于 SSVEP 的 BCI 系统表征疲劳行为,” 基于知识系统, 第 152 期, 第 163-171 页, 2018 年 7 月. [26] L. Cheng 等人, “动态功能连接的主状态揭示了静息状态和任务状态大脑之间的联系：一项 fMRI 研究,” 国际神经系统杂志, 第 28 卷, 第 7 期, 2018 年 4 月, 第 1850002 号文章. [27] H. Zhu 等人, “帕金森病中与皮层下网络相关的异常动态功能连接：从时间变化的角度看,” 前沿神经科学, 第 13 期, 第 80 页, 2019 年 2 月, doi: 10.3389/fnins.2019.00080. [28] D. Garrett, D. A. Peterson, C. W. Anderson 和 M. H. Thaut, “线性、非线性及特征选择方法在脑电图信号分类中的应用比较”，IEEE 神经系统与康复工程汇刊，第 11 卷，第 2 期，第 141-144 页，2003 年 6 月。[29]Z. K. Gao, W. D. Dang, M. X. Liu, W. Guo, K. Ma 和 G. R. Chen, “基于广泛学习的 VEP 脑机接口系统脑电图信号分类”，IEEE 系统、人与 Cybernetics 汇刊，早期访问，doi: 10.1109/TSMC.2020.2964684。[30]G. L. Ahern 和 G. E. Schwartz, “人类大脑中积极和消极情绪的差异侧化：脑电图频谱分析”，神经心理学杂志，第 23 卷，第 6 期，第 745-755 页，1985 年。[31]R. N. Duan, J. Y. Zhu 和 B. L. Lu, “基于脑电图的差分熵特征情绪分类”，IEEE/EMBS 神经会议论文集。

Eng., San Diego, CA, USA, 2013, pp. 81–84.

[32] S. Jirayucharoensak, S. Pan-Ngum, and P. Israsena, "EEG-based emotion recognition using deep learning network with principal component based covariate shift adaptation," *Sci. World J.*, vol. 2014, p. 10, Sep. 2014, doi: 10.1155/2014/627892.

[33] Y. H. Liu, C. T. Wu, W. T. Cheng, Y. T. Hsiao, P. M. Chen, and J. T. Teng, "Emotion recognition from single-trial EEG based on Kernel Fisher's emotion pattern and imbalanced quasiconformal kernel support vector machine," *Sensors*, vol. 14, no. 8, pp. 13361–13388, Aug, 2014.

[34] G. E. Hinton, S. Osindero, and Y. W. Teh, "A fast learning algorithm for deep belief nets," *Neural Comput.*, vol. 18, no. 7, pp. 1527–1554, Jul. 2006.

[35] C. Szegedy *et al.*, "Going deeper with convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Boston, MA, USA, 2015, pp. 1–9.

[36] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, 1997.

[37] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, Lake Tahoe, NV, USA, 2012, pp. 1097–1105.

[38] O. Abdel-Hamid, A. R. Mohamed, H. Jiang, L. Deng, G. Penn, and D. Yu, "Convolutional neural networks for speech recognition," *IEEE/ACM Trans. Audio, Speech, Language Process.*, vol. 22, no. 10, pp. 1533–1545, Oct. 2014.

[39] X. Shi *et al.*, "Deep learning for precipitation nowcasting: A benchmark and a new model," in *Proc. Adv. Neural Inf. Process. Syst.*, Long Beach, CA, USA, 2017, pp. 5622–5632.

[40] A. Alahi, K. Goel, V. Ramanathan, A. Robicquet, L. Fei-Fei, and S. Savarese, "Social LSTM: Human trajectory prediction in crowded spaces," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit*, Las Vegas, NV, USA, 2016, pp. 961–971.

[41] Y. R. Tabar and U. Halici, "A novel deep learning approach for classification of EEG motor imagery signals," *J. Neural Eng.*, vol. 14, no. 1, Feb. 2017, Art. no. 016003.

[42] T. Uktveris and V. Jusas, "Application of convolutional neural networks to four-class motor imagery classification problem," *Inf. Technol. Control*, vol. 46, no. 2, pp. 260–273, 2017.

[43] N. Lu, T. Li, X. Ren, and H. Miao, "A deep learning scheme for motor imagery classification based on restricted Boltzmann machines," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 25, no. 6, pp. 566–576, Jun. 2017.

[44] M. Hajinoroozi, Z. Mao, T.-P. Jung, C.-T. Lin, and Y. Huang, "EEG-based prediction of driver's cognitive performance by deep convolutional neural network," *Signal Process. Imag.*, vol. 47, pp. 549–555, Sep. 2016.

[45] Z. K. Gao *et al.*, "EEG-based spatio-temporal convolutional neural network for driver fatigue evaluation," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 9, pp. 2755–2763, Sep. 2019.

[46] Y. X. Yang *et al.*, "A recurrence quantification analysis-based channel-frequency convolutional neural network for emotion recognition from EEG," *Chaos*, vol. 28, no. 8, Aug. 2018, Art. no. 085724.

[47] J. P. Li, S. Qiu, C. D. Du, Y. X. Wang, and H. G. He, "Domain adaptation for EEG emotion recognition based on latent representation similarity," *IEEE Trans. Cogn. Develop. Syst.*, early access, doi: 10.1109/TCDS.2019.2949306.

[48] Y. M. Yang, Q. J. Wu, W.-L. Zheng, and B.-L. Lu, "EEG-based emotion recognition using hierarchical network with subnetwork nodes," *IEEE Trans. Cogn. Develop. Syst.*, vol. 10, no. 2, pp. 408–419, Jun. 2018.

[49] R. T. Schirrmeister *et al.*, "Deep learning with convolutional neural networks for EEG decoding and visualization," *Human Brain Map.*, vol. 38, no. 11, pp. 5391–5420, Nov. 2017.

[50] R. Jenke, A. Peer, and M. Buss, "Feature extraction and selection for emotion recognition from EEG," *IEEE Trans. Affect. Comput.*, vol. 5, no. 3, pp. 327–339, Jul.–Sep. 2014.

[51] J. Atkinson and D. Campos, "Improving BCI-based emotion recognition by combining EEG feature selection and kernel classifiers," *Expert. Syst. Appl.*, vol. 47, pp. 35–41, Apr. 2016.

[52] L.-C. Shi, Y.-Y. Jiao, and B.-L. Lu, "Differential entropy feature for EEG-based vigilance estimation," in *Proc. IEEE/EMBS Conf. Neural Eng. (EMBC)*, Osaka, Japan, 2013, pp. 6627–6630.

[53] W.-L. Zheng, W. Liu, Y. Lu, B.-L. Lu, and A. Cichocki, "EmotionMeter: A multimodal framework for recognizing human emotions," *IEEE Trans. Cybern.*, vol. 49, no. 3, pp. 1110–1122, Mar. 2019.

[54] W.-L. Zheng, J.-Y. Zhu, and B.-L. Lu, "Identifying stable patterns over time for emotion recognition from EEG," *IEEE Trans. Affect. Comput.*, vol. 10, no. 3, pp. 417–429, Jul.–Sep. 2019.

[55] S. M. Alarcao and M. J. Fonseca, "Emotions recognition using EEG signals: A survey," *IEEE Trans. Affect. Comput.*, vol. 10, no. 3, pp. 374–393, Jul.–Sep. 2019.

[56] V. J. Lawhern, A. J. Solon, N. R. Waytowich, S. M. Gordon, C. P. Hung, and B. J. Lance, "EEGNet: A compact convolutional neural network for EEG-based brain–computer interfaces," *J. Neural Eng.*, vol. 15, no. 5, 2018, Art. no. 056013.

[57] B. H. Kim and S. Jo, "Deep physiological affect network for the recognition of human emotions," *IEEE Trans. Affect. Comput.*, early access, doi: 10.1109/TAFFC.2018.2790939.

[58] R. G. Hefron, B. J. Borghetti, J. C. Christensen, and C. M. S. Kabban, "Deep long short-term memory structures model temporal dependencies improving cognitive workload estimation," *Pattern Recognit. Lett.*, vol. 94, pp. 96–104, Jul. 2017.

[59] P. Bashivan, I. Rish, M. Yeasin, and N. Codella, "Learning representations from EEG with deep recurrent-convolutional neural networks," in *Proc. Int. Conf. Learn. Represent.*, San Juan, Puerto Rico, 2016, pp. 1–15.

[60] T. Song, W. Zheng, P. Song, and Z. Cui, "EEG emotion recognition using dynamical graph convolutional neural networks," *IEEE Trans. Affect. Comput.*, early access, doi: 10.1109/TAFFC.2018.2817622.

[61] J. Li, Z. Zhang, and H. He, "Hierarchical convolutional neural networks for EEG-based emotion recognition," *Cognit. Comput.*, vol. 10, pp. 1–13, Dec. 2017.

[62] T. Zhang, W. Zheng, Z. Cui, Y. Zong, and Y. Li, "Spatial–temporal recurrent neural network for emotion recognition," *IEEE Trans. Cybern.*, vol. 49, no. 3, pp. 839–847, Mar. 2019.

[63] G. Huang, Z. Liu, K. Q. Weinberger, and L. van der Maaten, "Densely connected convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2017, pp. 4700–4708.

[64] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," in *Proc. Int. Conf. Mach. Learn.*, Lille, France, 2015, pp. 448–456.

[65] V. Nair and G. E. Hinton, "Rectified linear units improve restricted Boltzmann machines," in *Proc. Int. Conf. Mach. Learn.*, Haifa, Israel, 2000, pp. 807–814.

[66] D. M. Kline and V. L. Berardi, "Revisiting squared-error and cross-entropy functions for training neural network classifiers," *Neural Comput. Appl.*, vol. 14, no. 4, pp. 310–318, 2005.

[67] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. Int. Conf. Learn. Represent.*, San Diego, CA, USA, 2015, pp. 1–15.

[68] W.-L. Zheng and B.-L. Lu, "Investigating critical frequency bands and channels for EEG-based emotion recognition with deep neural networks," *IEEE Trans. Auton. Mental Develop.*, vol. 7, no. 3, pp. 162–175, Sep. 2015.

[69] S. B. Eysenck, H. J. Eysenck, and P. Barrett, "A revised version of the psychoticism scale," *Pers. Ind. Differ.*, vol. 6, no. 1, pp. 21–29, 1985.

[70] W. Liu, W. L. Zheng, and B. L. Lu, "Emotion recognition using multimodal deep learning," in *Proc. Int. Conf. Neural Inf. Process.*, Kyoto, Japan, 2016, pp. 521–529.

[71] Y. Li *et al.*, "Targeting EEG/LFP synchrony with neural nets," in *Proc. Adv. Neural Inf. Process. Syst.*, Long Beach, CA, USA, 2017, pp. 4623–4633.

[72] W. Zheng, "Multichannel EEG-based emotion recognition via group sparse canonical correlation analysis," *IEEE Trans. Cogn. Develop. Syst.*, vol. 9, no. 3, pp. 281–290, Sep. 2017.

[73] S. Keshmiri, H. Sumioka, J. Nakanishi, and H. Ishiguro, "Emotional state estimation using a modified gradient-based neural architecture with weighted estimates," in *Proc. IEEE Int. Joint Conf. Neural Netw.*, 2017, pp. 4371–4378.

[74] R. Gupta, K. U. R. Laghari, and T. H. Falk, "Relevance vector classifier decision fusion and EEG graph-theoretic features for automatic affective state characterization," *Neurocomputing*, vol. 174, pp. 875–884, Jan. 2016.

[75] X. Li, D. Song, P. Zhang, G. Yu, Y. Hou, and B. Hu, "Emotion recognition from multi-channel EEG data through convolutional recurrent neural network," in *Proc. IEEE Int. Conf. Bioinformat. Biomed. (BIBM)*, Guangdong, China, 2016, pp. 352–359.

[76] S. Tripathi, S. Acharya, R. D. Sharma, S. Mittal, and S. Bhattacharya, "Using deep and convolutional neural networks for accurate emotion classification on DEAP dataset," in *Proc. AAAI Conf. Innov. Appl.*, San Francisco, CA, USA, 2017, pp. 4746–4752.

[77] M. Ali, A. H. Mosa, F. Al Machot, and K. Kyamakya, "EEG-based emotion recognition approach for e-healthcare applications," in *Proc. IEEE Conf. Ubiquitous Future Netw.*, Vienna, Austria, 2016, pp. 946–950.

[78] H. Xu and K. N. Plataniotis, "Affective states classification using EEG and semi-supervised deep learning approaches," in *Proc. IEEE Workshop Multimedia Signal Process.*, 2016, pp. 1–6.

[32] S. Jirayucharoensak, S. Pan-Ngum, 和 P. Israsena, “基于深度学习网络的脑电图情绪识别，采用基于主成分的协变量偏移适应,” Sci. World J., 卷 2014, 第 10 页, 2014 年 9 月, doi: 10.1155/2014/627892. [33] Y. H. Liu, C. T. Wu, W. T. Cheng, Y. T. Hsiao, P. M. Chen, 和 J. T. Teng, “基于单次试验脑电图的情绪识别，采用核 Fisher 情绪模式和失衡准共形核支持向量机,” Sensors, 卷 14, 第 8 期, 第 13361–13388 页, 2014 年 8 月. [34] G. E. Hinton, S. Osindero, 和 Y. W. Teh, “深度信念网络的快速学习算法,” Neural Comput., 卷 18, 第 7 期, 第 1527–1554 页, 2006 年 7 月. [35] C. Szegedy 等, “卷积的深度探索,” 在 IEEE 计算机视觉与模式识别会议论文集中, 波士顿, MA, USA, 2015, 第 1–9 页. [36] S. Hochreiter 和 J. Schmidhuber, “长短期记忆网络,” Neural Comput., 卷 9, 第 8 期, 第 1735–1780 页, 1997. [37] A. Krizhevsky, I. Sutskever, 和 G. E. Hinton, “使用深度卷积神经网络的 ImageNet 分类,” 在 Advances in Neural Information Processing Systems 会议论文集中, 湖 Tahoe, NV, USA, 2012, 第 1097–1105 页. [38] O. Abdel-Hamid, A. R. Mohamed, H. Jiang, L. Deng, G. Penn 和 D. Yu, “用于语音识别的卷积神经网络,”IEEE/ACM 传输. 音频、语音、语言处理，vol. 22，no. 10，pp. 1533–1545，2014 年 10 月。[39] X. Shi 等人，“用于降水临近预报的深度学习：基准测试和一个新模型,”在处理系统进展的神经信息处理会议，长滩，CA，美国，2017 年，pp. 5622–5632。[40] A. Alahi、K. Goel、V. Ramanathan、A. Robicquet、L. Fei-Fei 和 S. Savarese，“社交 LSTM：拥挤空间中的人体轨迹预测,”在 IEEE 计算机视觉与模式识别会议，拉斯维加斯，NV，美国，2016 年，pp. 961–971。[41] Y. R. Tabar 和 U. Halici，“一种用于脑电图运动想象信号分类的新型深度学习方法,”神经工程杂志，vol. 14，no. 1，2017 年 2 月，文章编号 016003。[42] T. Uktveris 和 V. Jusas，“将卷积神经网络应用于四类运动想象分类问题,”信息技术控制，vol. 46，no. 2，pp. 260–273，2017 年。[43] N. Lu、T. Li、X. Ren 和 H. Miao，“基于限制玻尔兹曼机的运动想象分类深度学习方案,” IEEE 神经系统康复工程传输，vol. 25，no. 6，pp. 566–576，2017 年 6 月。[44] M. Hajinoroozi、Z. Mao、T.-P. Jung, C.-T. Lin, 和 Y. Huang, “基于 EEG 的深度卷积神经网络预测驾驶员认知性能,” 图像信号处理, 卷. 47, 页. 549–555, 2016 年 9 月. [45] Z. K. Gao 等人, “基于 EEG 的时空卷积神经网络用于驾驶员疲劳评估,” IEEE 神经网络与学习系统汇刊, 卷. 30, 第 9 期, 页. 2755–2763, 2019 年 9 月. [46] Y. X. Yang 等人, “基于递归量化分析的通道-频率卷积神经网络用于从 EEG 识别情绪,” 混沌, 卷. 28, 第 8 期, 2018 年 8 月, 文章编号. 085724. [47] J. P. Li, S. Qiu, C. D. Du, Y. X. Wang, 和 H. G. He, “基于潜在表示相似性的 EEG 情绪识别领域自适应,” IEEE 认知发展系统汇刊, 快速访问, doi: 10.1109/TCDS.2019.2949306. [48] Y. M. Yang, Q. J. Wu, W.-L. Zheng, 和 B.-L. Lu, “基于 EEG 的情绪识别使用具有子网络节点的分层网络,” IEEE 认知发展系统汇刊, 卷. 10, 第 2 期, 页. 408–419, 2018 年 6 月. [49] R. T. Schirrmeister 等人, “用于 EEG 解码和可视化的卷积神经网络深度学习,” 人类脑图谱, 卷. 38, 第 11 期, 页. 5391–5420, 2017 年 11 月. [50] R. Jenke, A. Peer, 和 M. Buss, “用于从脑电图识别情绪的特征提取与选择,” IEEE 情感计算汇刊, 第 5 卷, 第 3 期, pp. 327–339, 2014 年 7 月–9 月. [51] J. Atkinson 和 D. Campos, “通过结合脑电图特征选择和核分类器改进基于脑机接口的情绪识别,” 专家系统应用, 第 47 卷, pp. 35–41, 2016 年 4 月. [52] L.-C. Shi, Y.-Y. Jiao, 和 B.-L. Lu, “用于基于脑电图警觉度估计的差异熵特征,” 在 IEEE/EMBS 神经工程会议 (EMBC) 会议论文集中, 大阪, 日本, 2013 年, pp. 6627–6630. [53] W.-L. Zheng, W. Liu, Y. Lu, B.-L. Lu, 和 A. Cichocki, “EmotionMeter: 一种用于识别人类情绪的多模态框架,” IEEE 控制系统汇刊, 第 49 卷, 第 3 期, pp. 1110–1122, 2019 年 3 月. [54] W.-L. Zheng, J.-Y. Zhu, 和 B.-L. Lu, “识别用于从脑电图识别情绪的稳定时间模式,” IEEE 情感计算汇刊, 第 10 卷, 第 3 期, pp. 417–429, 2019 年 7 月–9 月. [55] S. M. Alarcao 和 M. J. Fonseca, “使用脑电图信号的情绪识别: 一篇综述,” IEEE 情感计算汇刊, 第 10 卷, 第 3 期, pp. 374–393, 2019 年 7 月–9 月. [56] V. J. Lawhern, A. J. Solon, N. R. Waytowich, S. M. Gordon, C. P. Hung 和 B. J. Lance, “EEGNet: 一种用于脑机接口的紧凑型卷积神经网络,” 神经工程杂志, 第 15 卷, 第 5 期, 2018 年, 第 056013 号文章. [57] B. H. Kim 和 S. Jo, “用于人类情绪识别的深度生理情感网络,” 情感计算 Transactions, 预发布, doi: 10.1109/TAFFC.2018.2790939. [58] R. G. Hefron, B. J. Borghetti, J. C. Christensen 和 C. M. S. Kabban, “深度长短期记忆结构模型时间依赖性, 改进认知负荷估计,” 模式识别快报, 第 94 卷, 第 96-104 页, 2017 年 7 月. [59] P. Bashivan, I. Rish, M. Yeasin 和 N. Codella, “使用深度循环卷积神经网络从 EEG 学习表示,” 在第 10 届国际学习表征会议论文集中, 波多黎各圣胡安, 2016 年, 第 1-15 页. [60] T. Song, W. Zheng, P. Song 和 Z. Cui, “使用动态图卷积神经网络的 EEG 情绪识别,” 情感计算 Transactions, 预发布, doi: 10.1109/TAFFC.2018.2817622. [61] J. Li, Z. Zhang 和 H. He, “用于基于 EEG 的情绪识别的层次卷积神经网络,” 认知 Comput., vol. 10, pp. 1–13, Dec. 2017. [62] 张涛, 郑伟, 崔峥, 宗宇, 和 李岩, “时空循环神经网络用于情绪识别,” IEEE 资讯科学汇, vol. 49, no. 3, pp. 839–847, Mar. 2019. [63] 黄钧, 刘铮, 魏凯昆, 和 van der Maaten 拉斯, “密集连接卷积网络,” 在 IEEE 计算机视觉与模式识别会议论文集, 2017, pp. 4700–4708. [64] Ioffe 斯坦利 和 Szegedy 卡罗尔, “批量归一化：通过减少内部协变量偏移来加速深度网络训练,” 在 国际机器学习会议论文集, 法国里尔, 2015, pp. 448–456. [65] Nair 维杰 和 Hinton 格雷·爱德华, “整流线性单元改进受限玻尔兹曼机,” 在 国际机器学习会议论文集, 以色列海法, 2000, pp. 807–814. [66] Kline 戴夫·M. 和 Berardi 维克多·L., “重新审视用于训练神经网络分类器的平方误差和交叉熵函数,” Neural Computing Applications, vol. 14, no. 4, pp. 310–318, 2005. [67] Kingma 德里克·P. 和 Ba 杰, “Adam: 一种随机优化的方法,” 在 学习表示国际会议论文集, 美国圣地亚哥, CA, 2015, pp. 1–15. [68] 郑伟立 和 郑伯伦 Lu，“基于深度神经网络研究脑电图情绪识别的关键频段和通道”，IEEE Transactions on Autonomous Mental Development，第 7 卷，第 3 期，第 162-175 页，2015 年 9 月。[69] S. B. Eysenck, H. J. Eysenck, 和 P. Barrett，“精神病量表修订版”，Personality and Individual Differences，第 6 卷，第 1 期，第 21-29 页，1985 年。[70] W. Liu, W. L. Zheng, 和 B. L. Lu，“基于多模态深度学习的情绪识别”，在神经信息处理国际会议论文集，日本京都，2016 年，第 521-529 页。[71] Y. Li 等人，“使用神经网络靶向脑电图/低频电位同步”，在先进神经信息处理系统会议论文集，美国加利福尼亚州长滩，2017 年，第 4623-4633 页。[72] W. Zheng，“基于多通道脑电图和组稀疏典型相关分析的情绪识别”，IEEE Transactions on Cognitive Development Systems，第 9 卷，第 3 期，第 281-290 页，2017 年 9 月。[73] S. Keshmiri, H. Sumioka, J. Nakanishi, 和 H. Ishiguro，“使用改进的基于梯度的加权估计神经架构进行情绪状态估计”，在 IEEE 国际联合神经网络会议论文集，2017 年，第 4371-4378 页。[74] R. Gupta, K. U. R. Laghari, 和 T. H. Falk，“相关向量分类器决策融合与脑电图图论特征用于自动情感状态表征,”神经计算，vol. 174，pp. 875–884，2016 年 1 月。[75] X. Li, D. Song, P. Zhang, G. Yu, Y. Hou 和 B. Hu，“基于多通道脑电图数据的卷积循环神经网络情绪识别,”在 IEEE 国际生物信息生物医学会议（BIBM）论文集中，中国广东，2016，pp. 352–359。[76] S. Tripathi, S. Acharya, R. D. Sharma, S. Mittal 和 S. Bhattacharya，“使用深度和卷积神经网络对 DEAP 数据集进行精确情绪分类,”在 AAAI 创新应用会议论文集中，美国加州旧金山，2017，pp. 4746–4752。[77] M. Ali, A. H. Mosa, F. Al Machot 和 K. Kyamakya，“基于脑电图的医疗保健应用情绪识别方法,”在 IEEE 普及未来网络会议论文集中，奥地利维也纳，2016，pp. 946–950。[78] H. Xu 和 K. N. Plataniotis，“使用脑电图和半监督深度学习方法进行情感状态分类,”在 IEEE 多媒体信号处理研讨会论文集中，2016，pp. 1–6。

[79] K. Ansari-Asl, G. Chanel, and T. Pun, "A channel selection method for EEG classification in emotion assessment based on synchronization likelihood," in *Proc. IEEE Eur. Signal Process. Conf.*, 2007, pp. 1241–1245.

[80] X. Jia, K. Li, X. Li, and A. Zhang, "A novel semi-supervised deep learning framework for affective state recognition on EEG signals," in *Proc. IEEE Int. Conf. Bioinformat. Bioeng.*, 2014, pp. 30–37.

Zhongke Gao (Senior Member, IEEE) received the M.Sc. and Ph.D. degrees from Tianjin University, Tianjin, China, in 2007 and 2010, respectively.

He is currently a Full Professor with the School of Electrical and Information Engineering, Tianjin University, where he is the Director of the Laboratory of Complex Networks and Intelligent Systems. His work has been cited more than 2600 times according to Google Scholar. His research interests include deep learning, EEG analysis, complex networks, brain–computer interface, and wearable intelligent devices. He has published more than 100 journal papers in the above fields.

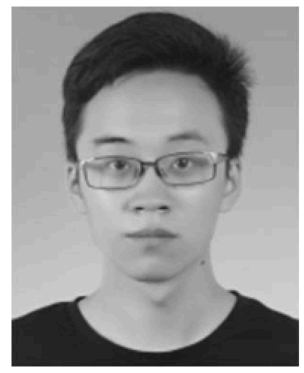
Prof. Gao has been serving as an Editorial Board Member for *Scientific Reports* and an Associate Editor of IEEE ACCESS, *Neural Processing Letters*, and *Royal Society Open Science*.

Xinmin Wang received the bachelor's degree in automation from the School of Electrical and Information Engineering, Tianjin University, Tianjin, China, in 2017, where he is currently pursuing the master's degree in control science and engineering with the School of Electrical and Information Engineering.

His research interests include affective computing, EEG analysis, and machine learning.

Yuxuan Yang received the bachelor's degree in automation from Anhui University, Hefei, China, in 2014, the master's degree in automation from the School of Electrical and Information Engineering, Tianjin University, Tianjin, China, in 2017, where she is currently pursuing the Ph.D. degree with the School of Electrical and Information Engineering.

Her research interests include affective computing, brain–computer interface, machine learning, and complex networks.



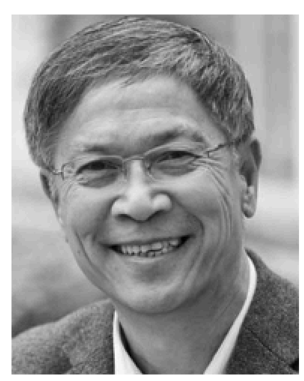
Yanli Li received the bachelor's degree in automation from the School of Electrical and Information Engineering, Tianjin University, Tianjin, China, in 2017, where he is currently pursuing the master's degree in control engineering with the School of Electrical and Information Engineering.

His research interests include affective computing, brain–computer interface, and machine learning.

Kai Ma received the Ph.D. degree from the University of Illinois at Chicago, Chicago, IL, USA, in 2014.

He worked with Siemens Medical Solution, Malvern, PA, USA, for more than five years. He is currently working as a Principal Researcher with Tencent, Shenzhen, China.

His research interests include medical image analysis, deep learning, computer vision, and brain–computer interface.

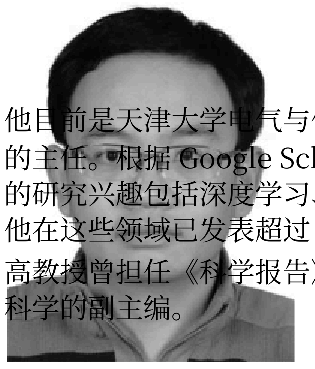


Guanrong Chen (Life Fellow, IEEE) received the M.Sc. degree in computer science from Sun Yat-sen University, Guangzhou, China, in 1981, and the Ph.D. degree in applied mathematics from Texas A&M University, College Station, TX, USA, in 1987.

He was a Tenured Full Professor with the University of Houston, Houston, TX, USA. Since 2000, he has been a Chair Professor and the Founding Director of the Center for Chaos and Complex Networks, City University of Hong Kong, Hong Kong.

Prof. Chen received the 2011 Euler Gold Medal in Russia and the conferred Honorary Doctorate by the Saint Petersburg State University, Russia, in 2011, and by the University of Le Havre, Romandie, France, in 2014. He is a Highly Cited Researcher in Engineering as well as in Mathematics according to Thomson Reuters. He is a member of the Academia of Europe and a fellow of the World Academy of Sciences.

[79] K. Ansari-Asl, G. Chanel, 和 T. Pun, “基于同步可能性的脑电图分类情绪评估的通道选择方法,” 在《IEEE 欧洲信号处理会议论文集》中, 2007, 第 1241–1245 页。 [80] X. Jia, K. Li, X. Li, 和 A. Zhang, “一种用于脑电信号情绪状态识别的新型半监督深度学习框架,” 在《IEEE 国际生物信息与生物工程会议论文集》中, 2014, 第 30–37 页。



Zhongke Gao（IEEE 高级会员）于 2007 年和 2010 年分别从中国天津大学获得理学硕士学位和博士学位。

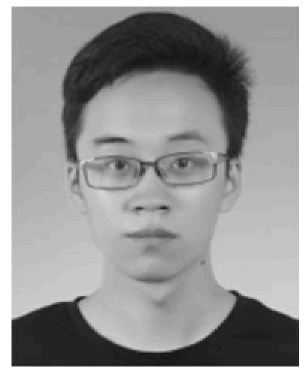
他目前是天津大学电气与信息工程学院的正教授，也是复杂网络与智能系统实验室的主任。根据 Google Scholar 的数据，他的研究成果已被引用超过 2600 次。他的研究兴趣包括深度学习、脑电图分析、复杂网络、脑机接口和可穿戴智能设备。他在这些领域已发表超过 100 篇期刊论文。

高教授曾担任《科学报告》的编委，以及 IEEE A、神经处理快报和皇家学会开放科学的副主编。

王新民于 2017 年获得天津大学电气与信息工程学院自动化专业的学士学位，目前在该学院攻读控制科学与工程硕士学位。他的研究兴趣包括情感计算、脑电图分析和机器学习。

杨宇轩于 2014 年获得中国合肥安徽大学的自动化学士学位，2017 年获得中国天津天津大学电气与信息工程学院的自动化硕士学位，目前在该学院攻读博士学位。

她的研究兴趣包括情感计算、脑机接口、机器学习和复杂网络。

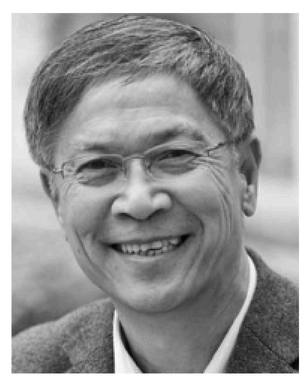


李艳丽于 2017 年获得中国天津天津大学电气与信息工程学院的自动化学士学位，目前在该学院攻读控制工程硕士学位。他的研究兴趣包括情感计算、脑机接口和机器学习。

马凯于 2014 年获得美国伊利诺伊大学芝加哥分校的博士学位。

他在美国宾夕法尼亚州马尔文市的西门子医疗解决方案公司工作了五年多。他目前在中国深圳的腾讯公司担任首席研究员。

他的研究兴趣包括医学图像分析、深度学习、计算机视觉和脑机接口。



陈关荣（IEEE 终身 Fellow）于 1981 年在中国广州的中山大学获得计算机科学硕士学位，并于 1987 年在美国德克萨斯州阿马尔回复大学的科林斯敦分校获得应用数学博士学位。

他曾是美国德克萨斯州休斯顿市的休斯顿大学终身正教授。自 2000 年以来，他一直是香港城市大学混沌与复杂网络中心主任教授和创始主任。

香港。陈教授于 2011 年在俄罗斯获得欧拉金质奖章，并于 2011 年由俄罗斯圣彼得堡国立大学授予名誉博士学位，2014 年由法国勒阿弗尔大学授予名誉博士学位。根据汤森路透的数据，他是一位在工程和数学领域的高被引学者。他是欧洲科学院的成员，也是世界科学院的院士。